



MODERN INSTITUTE OF TECHNOLOGY & RESEARCH CENTRE, ALWAR

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Subject & Subject Code: BEE, 2FY2-07

Semester: II

Session: 2022-23 (Even Sem)

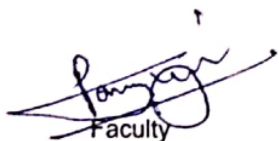
Branch: First Year

Batch: D

Unit: IV

Date of Submission: 21/03/23

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Faculty


Coordinator


Dean Academics

UNIT-4

LECTURE NO. 1

BJT CIRCUITS

BJT INTRODUCTION & CONSTRUCTION

INTRODUCTION :

Transistors are one of the very important components used in electronic circuit constructions. These humble components can almost be found everywhere; from simple relay driver circuits to complex motherboard circuits, transistors prove their presence. In fact, your microcontrollers and microprocessors are nothing but a collection of a large number of transistors synthesized to perform a collective operation.

Do remember that many switching devices like BJT, MOSFET, IGBT, SCR, TRIAC, DIAC, etc. can be collectively called transistors. But, the most basic (oldest) of the transistor is the BJT Transistor, hence in this article, we get into details of that, you can use the links to learn more about other power switches.

BJT is the short form of Bipolar Junction Transistor, it is a solid-state current-controlled device which can be used to electronically switch a circuit, you can think of it as your normal Fan or Light switch, but instead of you turning it on manually it can be controlled electronically. Technically speaking, BJT is a three-terminal device with an Emitter, collector, and a base pin, the current flow through the emitter and collector are controlled by the amount of current applied to the base.

Transistor is device which is made up of Transfer + Resistor means it transfer the Resistance from Input Side to Output Side

BJT stands for Bipolar Junction Transistor.

Bipolar means in it current flow due to two types of charge carriers i.e Holes & Electrons.

It is three terminal , two junction bipolar device which is current controlled used for many applications like Switching, Amplification, Impedance Matching etc.

The word Transistor is a combination of the two words Transfer Varistor which describes their mode of operation way back in their early days of electronics development.

There are two basic types of bipolar transistor construction, PNP and NPN, which basically describes the physical arrangement of the P-type and N-type semiconductor materials from which they are made.

The Bipolar Transistor basic construction consists of two PN-junctions producing three connecting terminals with each terminal being given a name to identify it from the other two. These three terminals are known and labelled as the Emitter (E), the Base (B) and the Collector (C) respectively.

Bipolar Transistors are current regulating devices that control the amount of current flowing through them from the Emitter to the Collector terminals in proportion to the amount of biasing voltage applied to their base terminal, thus acting like a current-controlled switch.

As a small current flowing into the base terminal controls a much larger collector current forming the basis of transistor action.

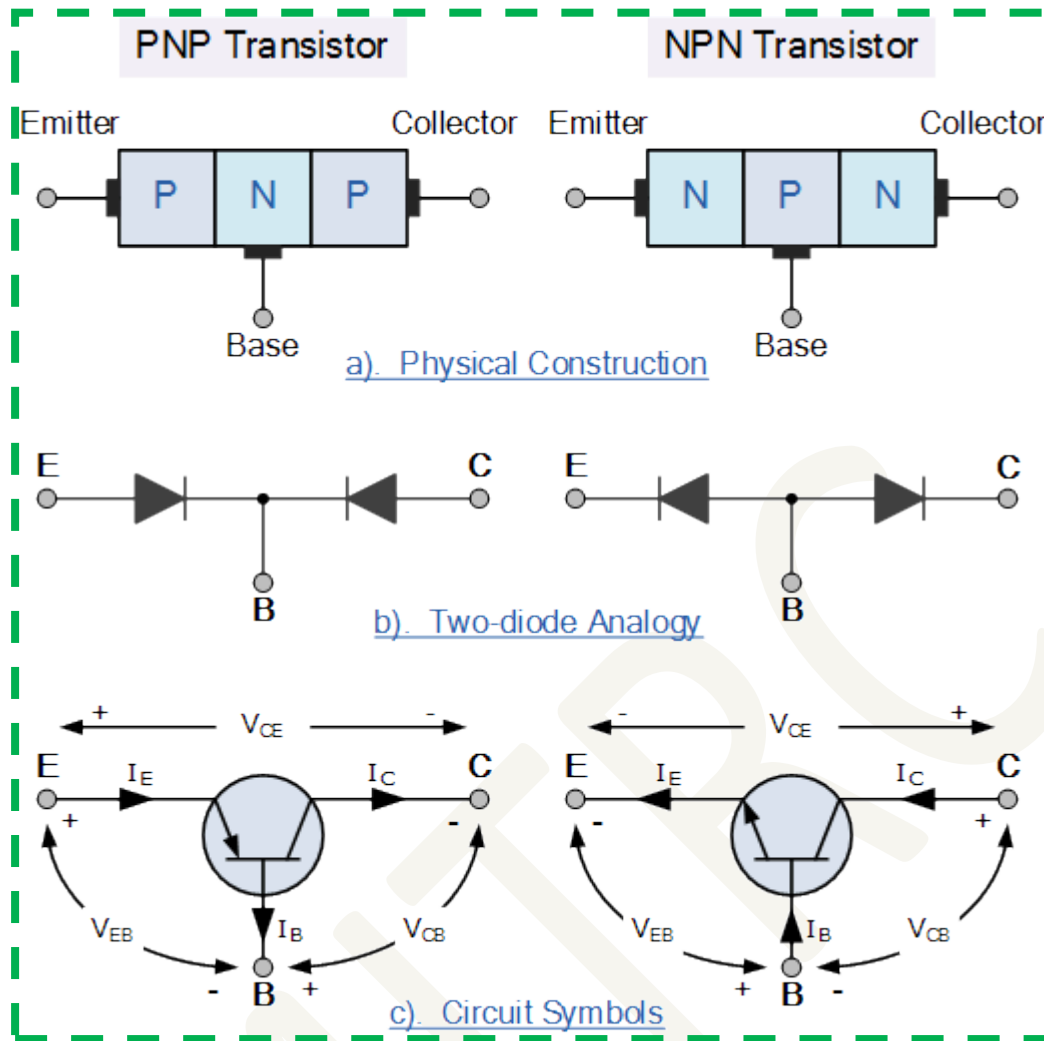


Figure 1: BJT Physical Construction ,Symbol, Construction

CONSTRUCTION :

BJT is three layer device. The three layers are Emitter , Base & Collector respectively.

Emitter : It is the terminal which emit large number of charge carriers . In NPN it emit electrons & in PNP it emit Holes. It forms the Emitter Current .

To emit large amount of charge carriers , it should be heavily doped . Area is moderate .

Base : It is Controlling Terminal. In it Charge Carriers that are emitted by Emitter recombines with opposite type of charge carriers.

Base is Thin & Lightly Doped so that recombination reduced .

Collector : It is moderate doped & area is large. It collect the charge carriers that are emitted by Emitter after recombination in Base.

Since Collector Base Junction is Reverse Biased so Heat dissipated. To protect the BJT , Collector area should be large so that heat distributed in Large Area.

Emitter & Base form Junction E-B

Collector & Base form C-B Junction

Since Emitter is heavily doped so the depletion layer is narrow in Emitter. Than it is moderate in Collector. In Base it is more Wider. The Area remained free from Depletion Layer in Base is called Effective Base Width.

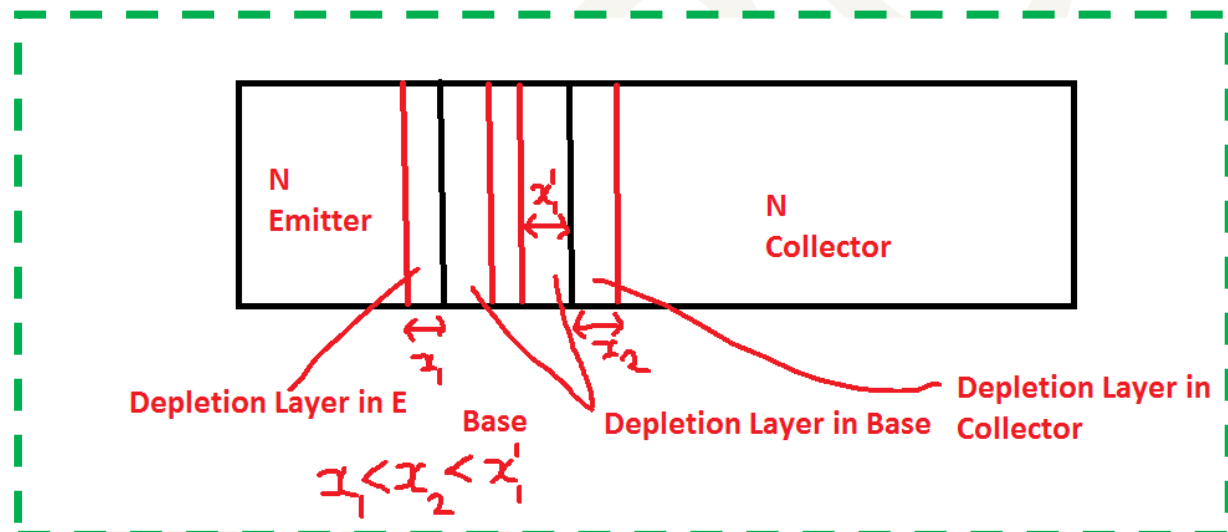


Figure 2: BJT ARCHITECTURE

The construction and circuit symbols for both the PNP and NPN bipolar transistor are given above with the arrow in the circuit symbol always showing the direction of “conventional current flow” between the base terminal and its emitter terminal.

The direction of the arrow always points from the positive P-type region to the negative N-type region for both transistor types, exactly the same as for the standard diode symbol.

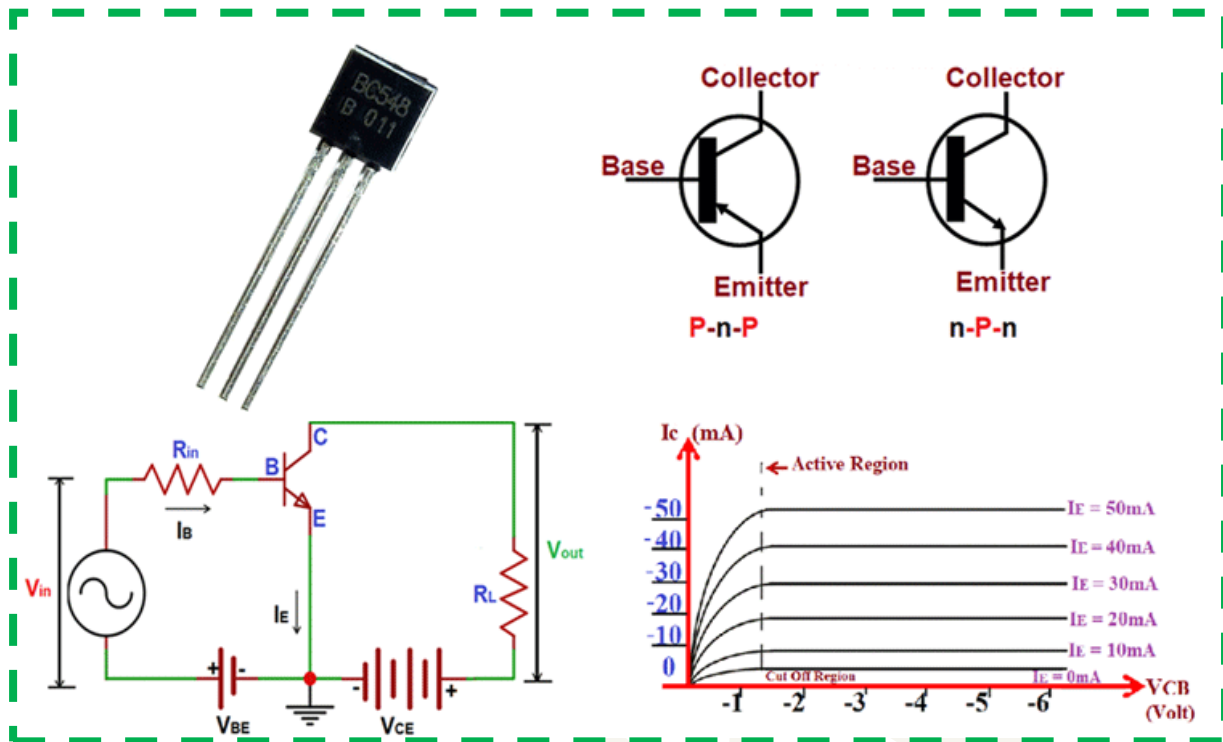


Figure 3: BJT PHYSICAL, SYMBOL, BJT I-V MEASURE CIRCUIT, CHARECTERISTICS

The difference between the PNP and NPN transistors is that the arrow mark at the emitter end if you have noticed, the arrow in the PNP transistor is mentioned as moving from the emitter to the base whereas in the NPN transistor the arrow will be moving from the base to the emitter.

The Direction of the arrow represents the direction of current flow in the transistor, in PNP the current will be flowing from emitter to base, similarly in the NPN transistor current will be flowing from the base to emitter.

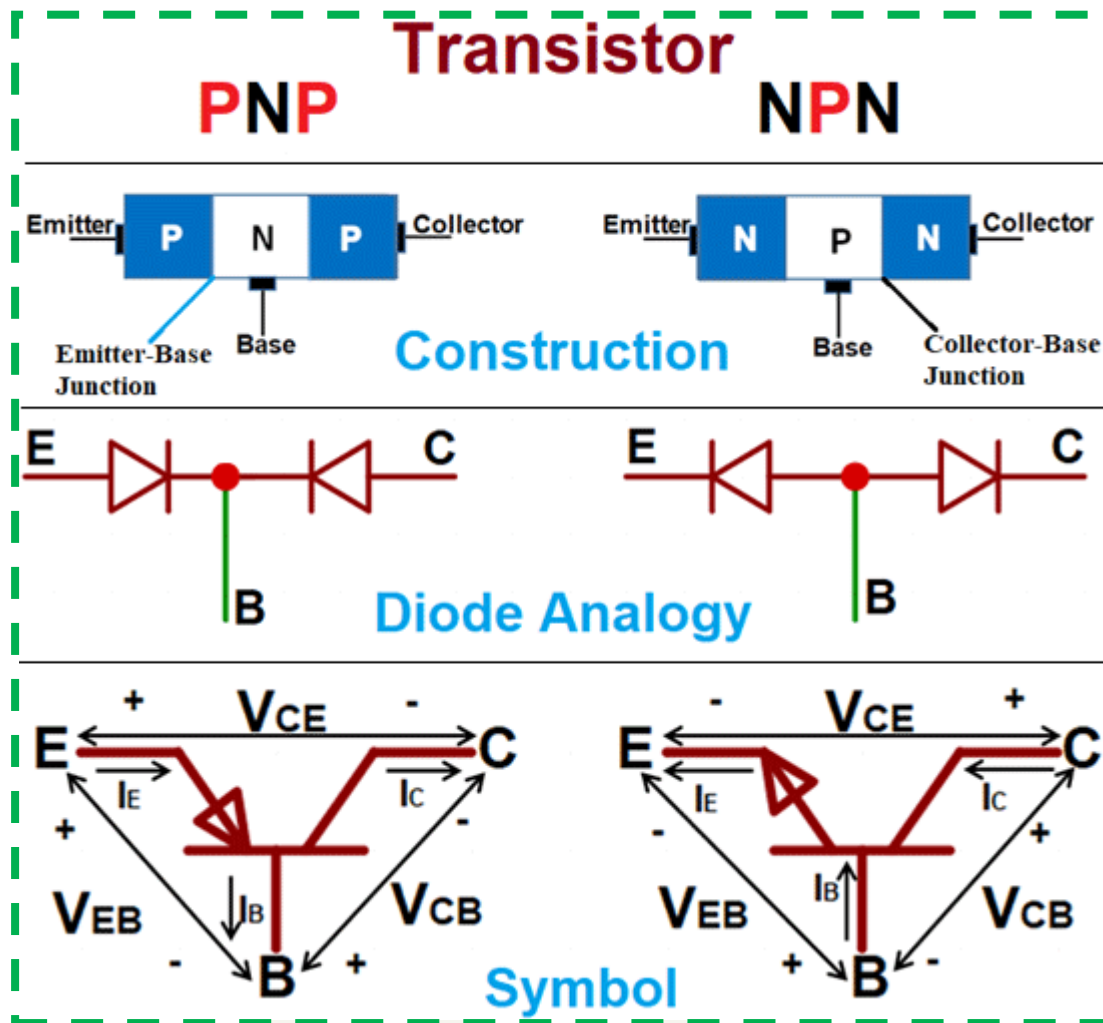


Figure 4: BJT PHYSICAL, SYMBOL, BJT I-V MEASURE CIRCUIT, CHARECTERISTICS

UNIT 4

BJT CIRCUITS

LECTURE NO. 2

BJT CIRCUITS

BJT WORKING

WORKING OF BJT :

The DC supply is provided for the operation of a transistor. This DC supply is given to the two PN junctions of a transistor which influences the actions of majority carriers in these emitter and collector junctions.

The junctions are forward biased and reverse biased based on our requirement. Forward biased is the condition where a positive voltage is applied to the p-type and negative voltage is applied to the n-type material. Reverse biased is the condition where a positive voltage is applied to the n-type and negative voltage is applied to the p-type material.

BJT operate in four regions

The supply of suitable external dc voltage is called as biasing. Either forward or reverse biasing is done to the emitter and collector junctions of the transistor. These biasing methods make the transistor circuit to work in four kinds of regions such as Active region, Saturation region, Cutoff region and Inverse active region (seldom used) . This is understood by having a look at the following table.

EMITTER JUNCTION	COLLECTOR JUNCTION	REGION OF OPERATION
Forward biased	Forward biased	Saturation region
Forward biased	Reverse biased	Active region
Reverse biased	Forward biased	Inverse active region
Reverse biased	Reverse biased	Cutoff region

Active region

This is the region in which transistors have many applications. This is also called as linear region. A transistor while in this region, acts better as an Amplifier.

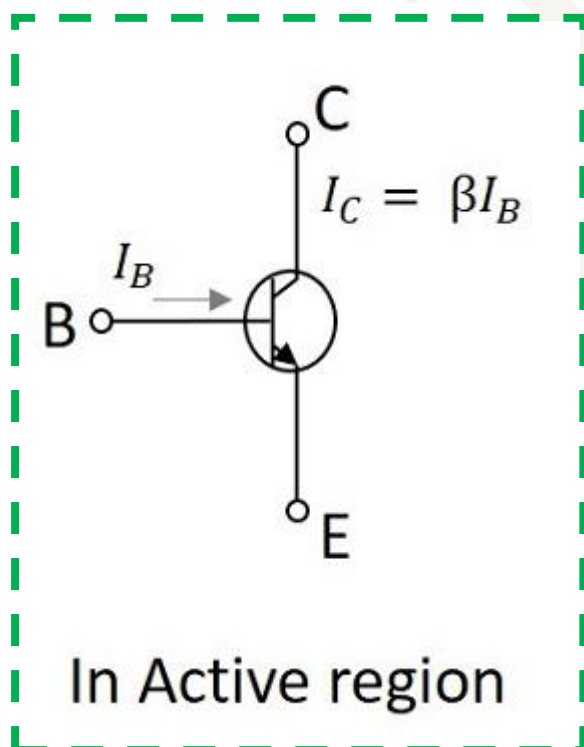


Figure 5: ACTIVE REGION

This region lies between saturation and cutoff. The transistor operates in active region when the emitter junction is forward biased and collector junction is reverse biased. In the active state, collector current is β times the base current, i.e.,

Saturation region

This is the region in which transistor tends to behave as a closed switch. The transistor has the effect of its collector and Emitter being shorted. The collector and Emitter currents are maximum in this mode of operation.

The figure below shows a transistor working in saturation region.

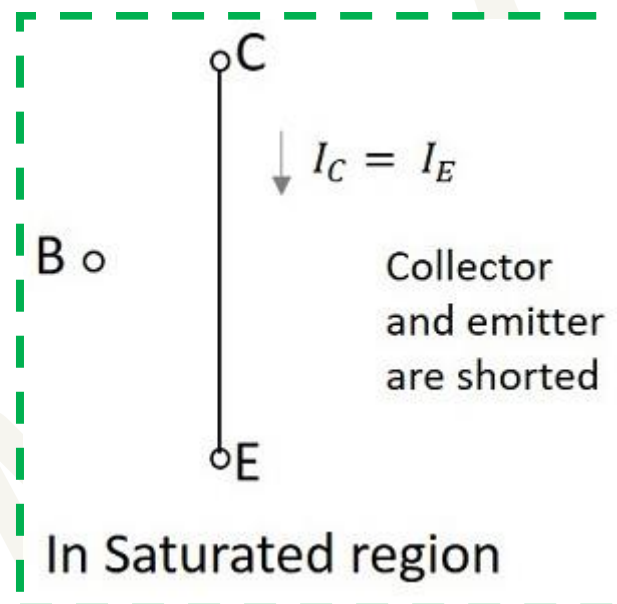


Figure 6: SATURATION REGION

The transistor operates in saturation region when both the emitter and collector junctions are forward biased. As it is understood that, in the saturation region the transistor tends to behave as a closed switch.

Cutoff region

This is the region in which transistor tends to behave as an open switch. The transistor has the effect of its collector and base being opened. The collector, emitter and base currents are all zero in this mode of operation.

The following figure shows a transistor working in cutoff region.

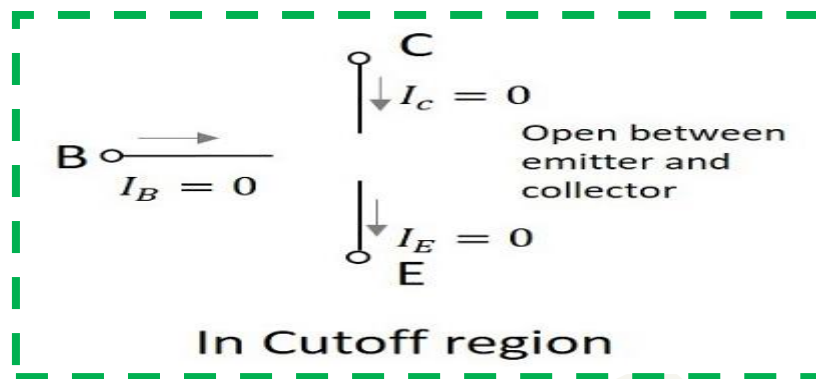


Figure 8: CUT OFF REGION

The transistor operates in cutoff region when both the emitter and collector junctions are reverse biased. As in cutoff region, the collector current, emitter current and base currents are nil.

ACTIVE REGION :

When Emitter Base Junction is Forward Biased than at particular voltage of $V_{EB} = 0.7$ Volt for Si depletion layer get vanish at Emitter-Base Junction . So majority Charge carriers let Electrons for NPN emitted in to Base . Similiarly the minority charge carriers of Base enter in to Emitter . So a Emitter Current flow due to these charge carriers crossing the E-B Junction .

Let 100 Charge Carriers i.e Electrons emitted from Emitter .

Now the Charge Carriers emitted from Emitter recombine with the Minority Charge Carrier in Base . Let 2% Recombine with Base Minority Charge Carriers. So Minority Charge Carriers reduced . To fulfil this Battery Provide the holes to Base . As a result Base Current flow.

Now Remaining 98% charge carriers cross the C-B Junction due to V_{CB} . Since Collector base Junction is Reverse Biased but Charge Carriers of Emitters are acting as Minority Carriers in Base . So they cross the Junction

As a result Collector Current Flow.

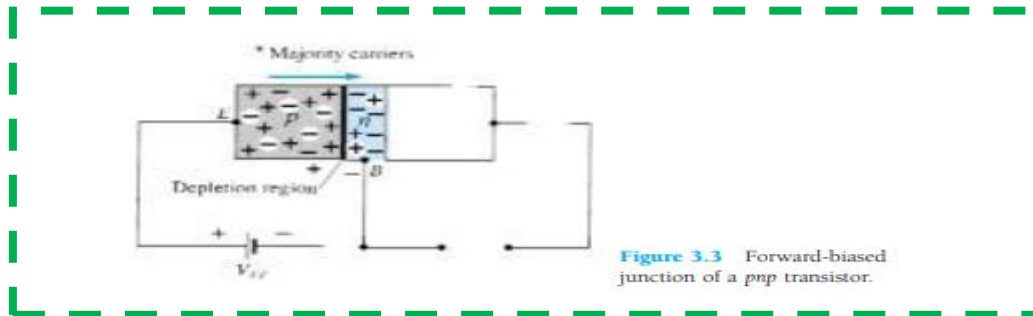


Figure 9: EMITTER BASE JUNCTION FORWARD BIASED

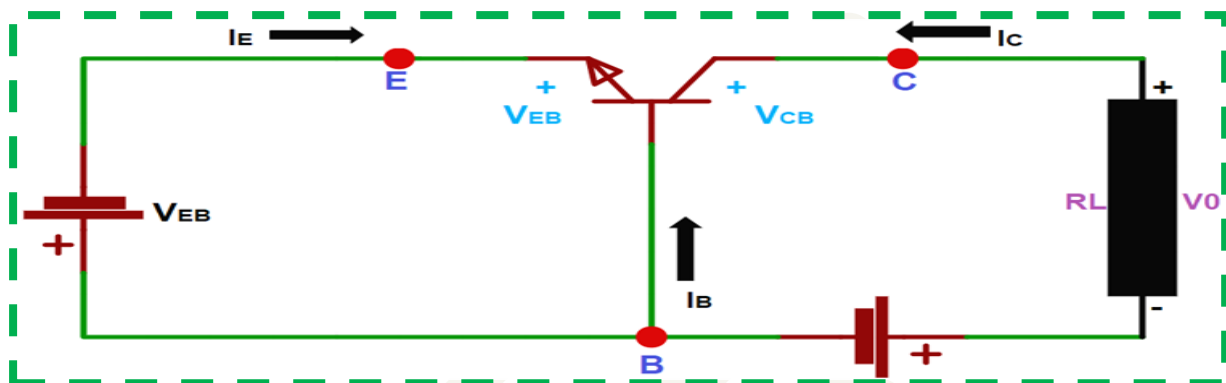
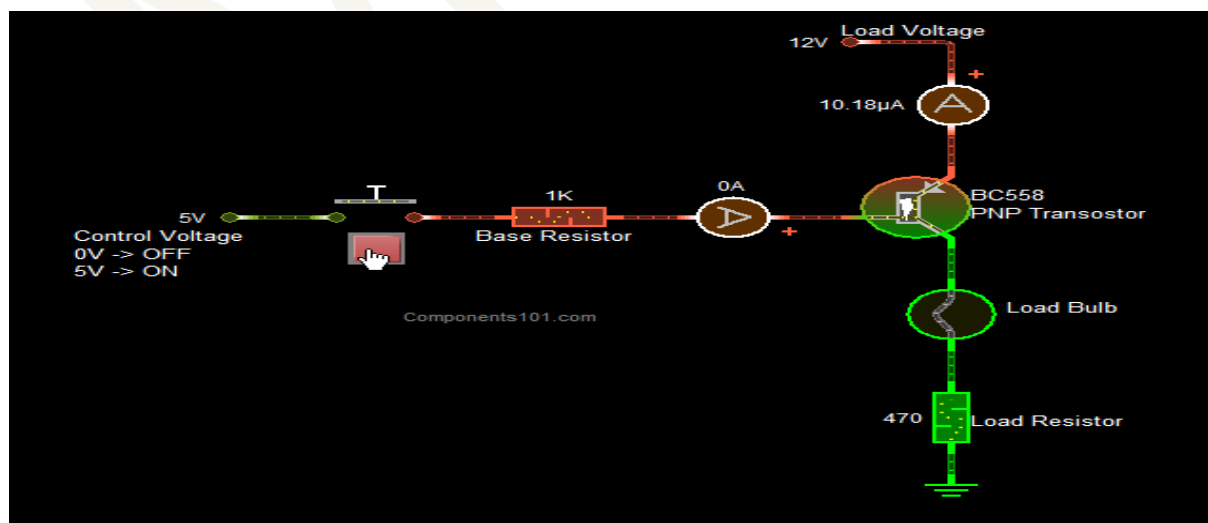


Figure 10: BJT IN ACTIVE REGION



UNIT 4

LECTURE NO. 3

BJT CIRCUITS

CURRENT COMPONENT

CURRENT COMPONENT OF BJT :

The conduction of current in NPN transistor is owing to electrons and in PNP transistor, it is owing to holes. The direction of current flow will be in opposite direction. Here, we can discuss the current components in a PNP transistor with common base configuration.

The emitter-base junction (JE) is forward biased and the collector-base junction (JC) is reversed biased as shown in figure. All the current components related to this transistor are shown here.

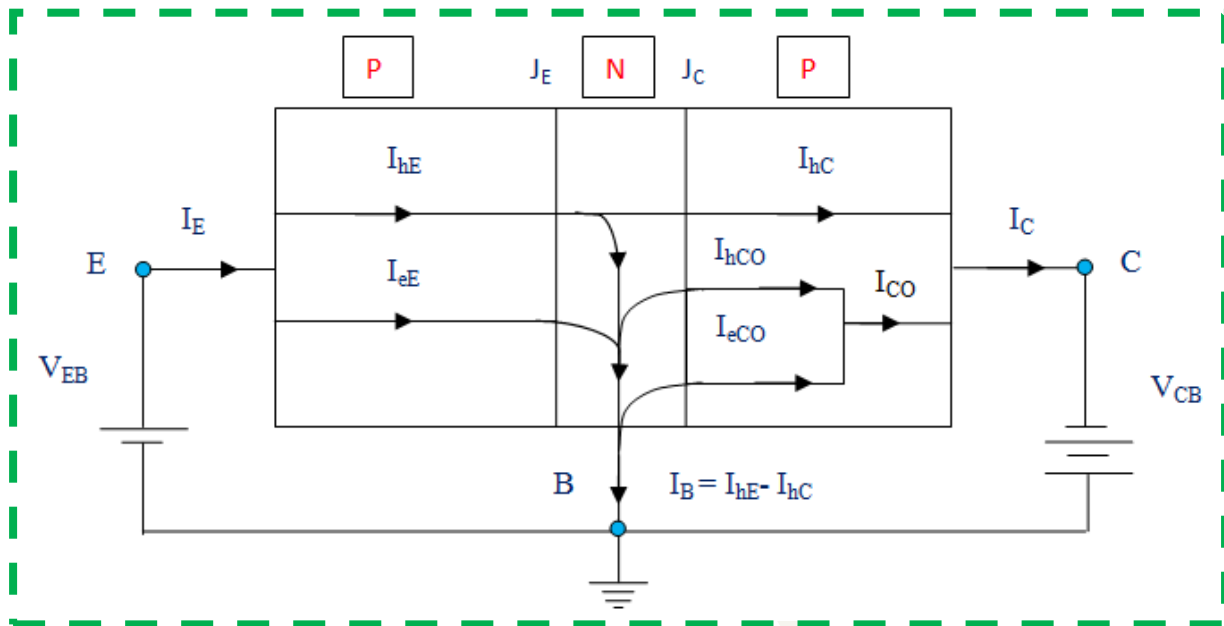


Figure 11: BJT CURRENT COMPONENT IN ACTIVE REGION

We know that, the current arrives the transistor through the emitter and this current is called emitter current (I_E). This current consists of two constituents – Hole current (I_{hE}) and Electron current (I_{eE}). I_{eE} is due to passage of electrons from base to emitter and I_{hE} is due to passage of holes from emitter to base.

$$I_E = I_{hE} + I_{eE}$$

Normally, the emitter is heavily doped compared to base in industrial transistor. So, the Electron current is negligible compared to Hole current. Thus we can conclude that, the whole emitter current in this transistor is due to the passage of holes from the emitter to the base.

Some of the holes which are crossing the junction J_E (emitter junction) combines with the electrons present in the base (N-type). Thus, every holes crossing J_E will not arrive at J_C . The remaining holes will reach the collector junction which produces the hole current component, I_{hC} . There will be bulk recombination in the base and the current leaving the base will be

There will be bulk recombination in the base and the current leaving the base will be

$$I_B = I_{hE} - I_{hC}$$

The electrons in the base which are lost by the recombination with holes (injected into the base across J_E) are refilled by the electrons that enter into the base region. The holes which are arriving at the collector junction (J_C) will cross the junction and it will go into the collector region.

When the emitter circuit is open circuited, then $I_E = 0$ and $I_{hC} = 0$. In this condition, the base and collector will perform as reverse biased diode. Here, the collector current, I_C will be same as reverse saturation current (I_{CO} or I_{CBO}). I_{CO} is in fact a small reverse current which passes through the PN junction diode. This is due to thermally generated minority carriers which are pushed by barrier potential. This reverse current increase; if the junction is reverse biased and it will have the same direction as the collector current. This current attains a saturation value (I_0) at moderate reverse biased voltage.

When the emitter junction is at forward biased (in active operation region), then the collector current will become

$$I_C = \alpha I_E + I_{CO}$$

The α is the large signal current gain which is a fraction of the emitter current which comprises of I_{hC} .

When the emitter is at closed condition, then $I_E \neq 0$ and collector current will be

$$I_C = I_{CO} + I_{hC}$$

In a PNP transistor, the reverse saturation current (I_{CBO}) will comprises of the current due to the holes passing through the collector junction from the base to collector region (I_{hCO}) and the current due to the electrons which are passing through the collector junction in the opposite direction (I_{eCO}).

$$\text{Therefore, } I_{CO} = I_{hCO} + I_{eCO}$$

The total current entering into the transistor will be equal to the total current leaving the transistor (according to Kirchhoff's current law).

$$\text{So, } I_E = I_C + I_B \text{ or } I_E = -(I_C + I_B)$$

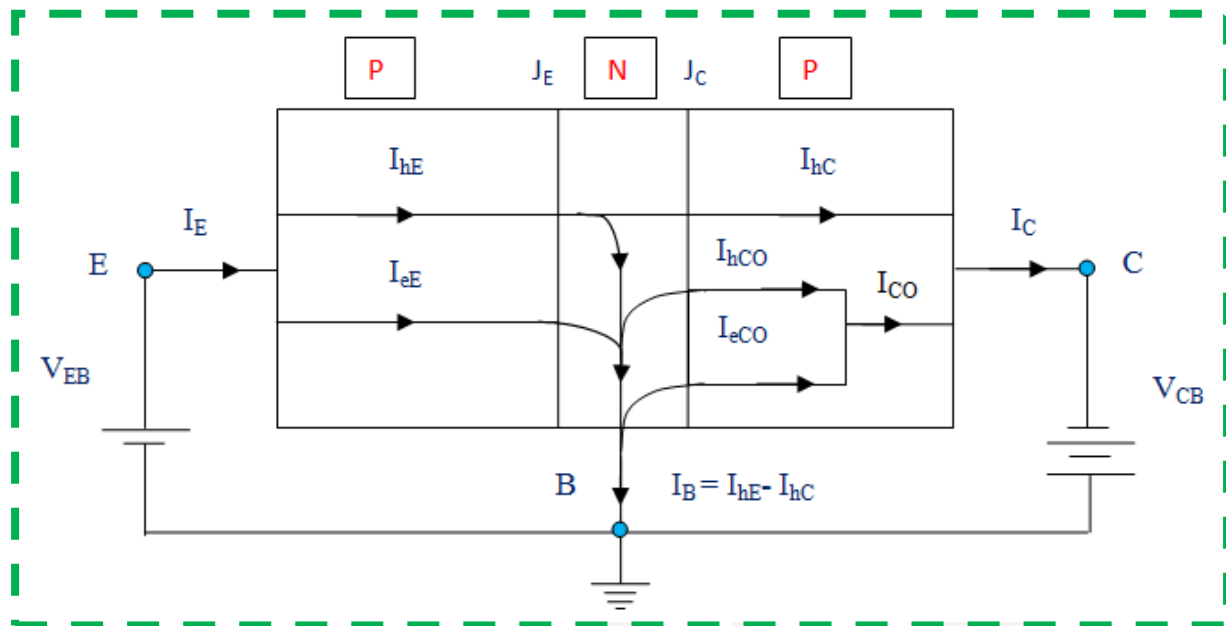


Figure 11: BJT CURRENT COMPONENT IN ACTIVE REGION

UNIT 4

LECTURE NO. 4

BJT CIRCUITS

CHARACTERISTICS OF BJT

I-V CURVE OF BJT :

Bipolar Transistor Configurations

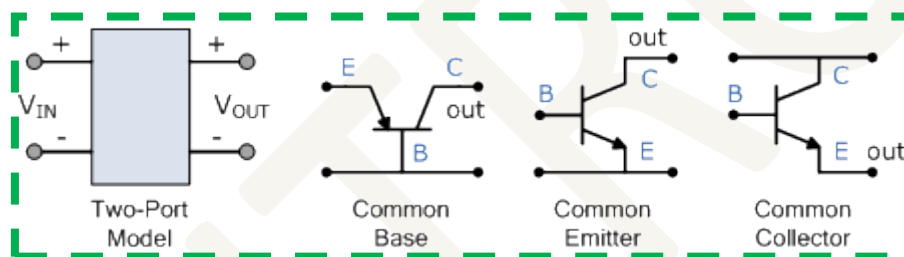


Figure 12: BJT CONFIGURATION

In two port Network two terminals are for input & two terminals are for Output . But in BJT there are three terminals so we use one terminal as Common for both Input & Output.

On the basis of this there are three configurations

Characteristic	Common Base	Common Emitter	Common Collector
Input Impedance	Low	Medium	High

Output Impedance	Very High	High	Low
Phase Shift	0°	180°	0°
Voltage Gain	High	Medium	Low
Current Gain	Low	Medium	High
Power Gain	Low	Very High	Medium

- Common Base Configuration
- Common Emitter Configuration
- Common Collector Configuration

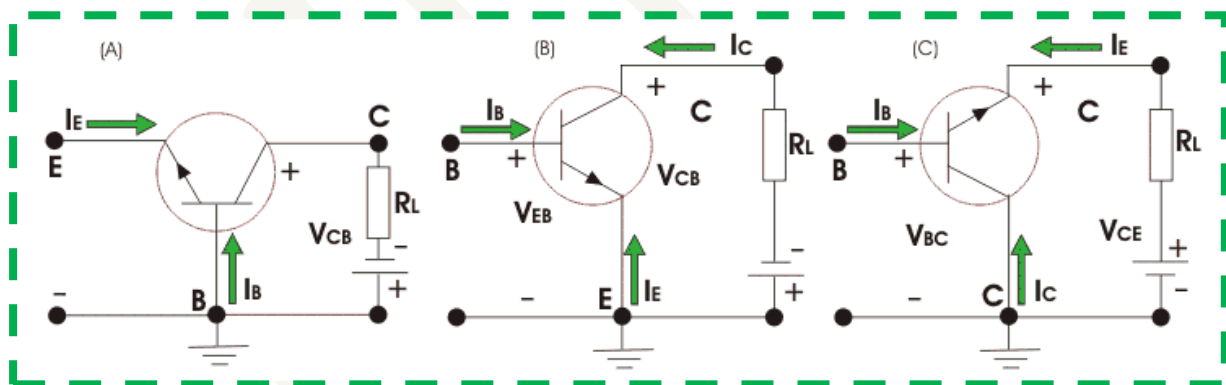


Figure 13: BJT CONFIGURATION CIRCUITS

Now coming to the characteristics of BJT there are different characteristics for different modes of operation. Characteristics is nothing but the graphical forms of relationships among different current and voltage variables of the transistor. The characteristics for p-n-p transistors are given for different modes and different parameters.

Characteristics Curve is of 2 types in BJT :

- Input Characteristics Curve
- Output Characteristics Curve

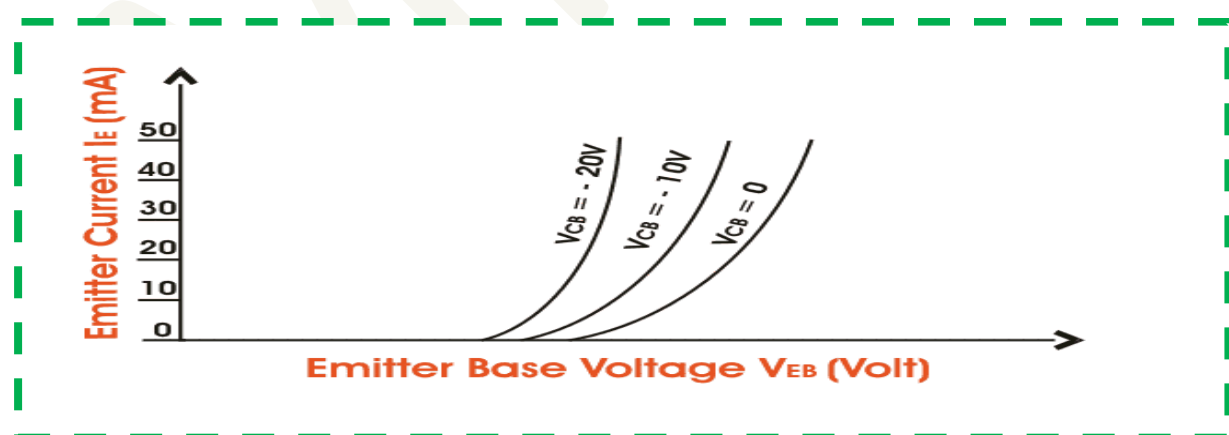
Input Curve is plot of Input Voltage & Input Current keeping Output Voltage Current Constant .

Output Curve is plot of Output Voltage & Output Current keeping Input Current Constant .

Common Base Characteristics

For p-n-p transistor, the input current is the emitter current (I_E) and the input voltage is the collector base voltage (V_{CB}). When we applied the forward bias voltage than E-B Junction acts as forward biased diode than a current will flow by diode's current relation . So at Cut in Voltage .7V for Si Current start to flow & rises exponentially with input Voltage .

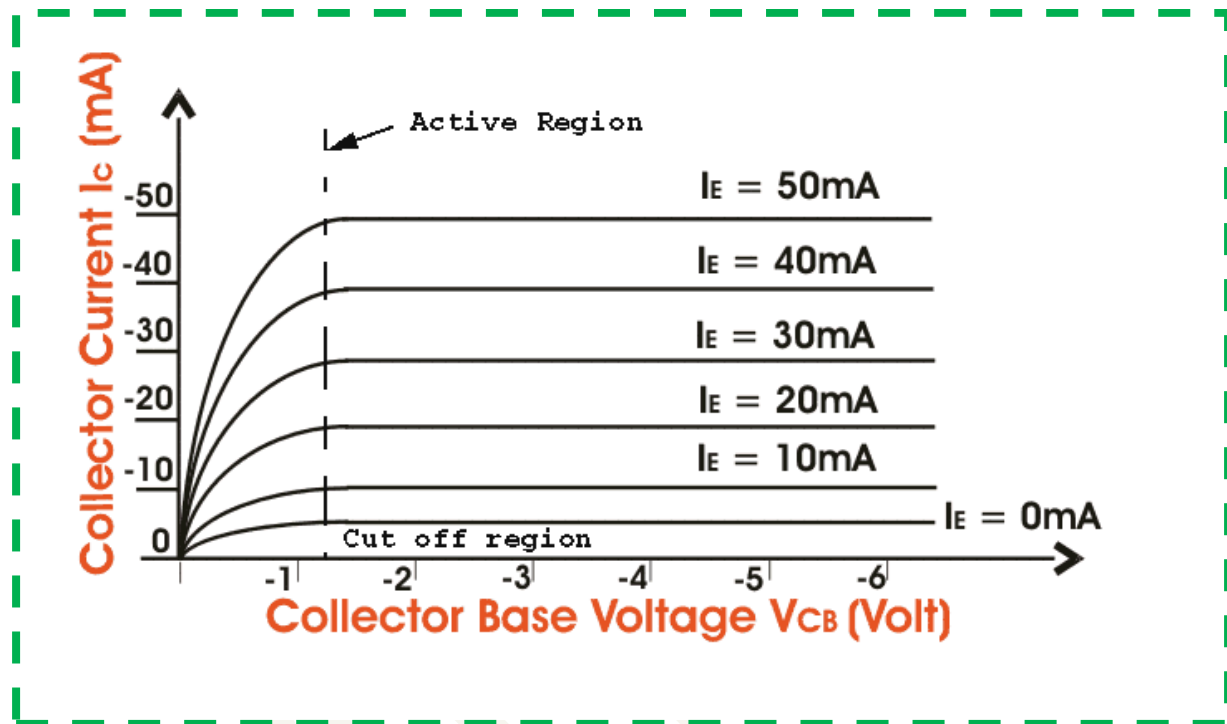
If we increase the Output Voltage than the depletion layer width of C-B Junction increase. So Effective Base Width decrease & Recombination reduces . It increase the Current Gain & As a Result Emitter Current Increases . It is Called Early Effect.



As the emitter-base junction is forward biased, therefore the graph of I_E Vs V_{EB} is similar to the forward characteristics of a p-n diode. I_E increases for fixed V_{EB} when V_{CB} increases.

Output Characteristics

The output characteristics show the relation between the output voltage and output current I_C is the output current and collector-base voltage and the emitter current I_E is the input current and works as the parameters. The figure below shows the output characteristics for a p-n-p transistor in CB mode.



:

As we know for p-n-p transistors I_E and V_{EB} are positive and I_C , I_B , V_{CB} are negative. These are three regions in the curve, active region saturation region and the cut off region. The active region is the region where the transistor operates normally.

Here the emitter junction is reverse biased. Now the saturation region is the region where both the emitter-collector junctions are forward biased. And finally, the cut off region is the region where both emitter and the collector junctions are reverse biased.

The active region is defined by the biasing arrangements of Fig. 3.6. At the lower end of the active region the emitter current (I_E) is zero, the collector current is simply that due to the reverse saturation current I_{CO} , as indicated in Fig.

The current I_{CO} is so small (microamperes) in magnitude compared to the vertical scale of I_C (milliamperes) that it appears on virtually the same horizontal line as I_{C0} . The circuit conditions that exist when $I_E = 0$ for the common-base configuration.

The notation most frequently used for I_{CO} on data and specification sheets is, as indicated in Fig, I_{CBO} . Because of improved construction techniques, the level of I_{CBO} for general-purpose transistors (especially silicon) in the low- and midpower ranges is usually so low that its effect can be ignored.

However, for higher power units I_{CBO} will still appear in the microampere range. In addition, keep in mind that I_{CBO} , like I_S for the diode (both reverse leakage currents) is temperature sensitive. At higher temperatures the effect of I_{CBO} may become an important factor since it increases so rapidly with temperature.

As in Active Region if we increase the Output Voltage then Collector Current rises linearly simultaneously depletion layer also spread. So Recombination reduce. At a particular condition the Collector Current = Emitter Current means $\alpha = 1$.

UNIT 4

LECTURE NO. 5

BJT CIRCUITS

BJT APPLICATIONS

BJT AS SWITCH :

If BJT operate in Saturation & Cut off than it act as Switch.

In Saturation , Maximum Current flow through Collector & Output Voltage= 0.2V

In Cut Off , Collector Current= 0 & Output Voltage = Maximum (V_{CC})

A switch offers open circuit (infinite resistance) when it is put in 'OFF' position and offers short circuit (zero resistance) when it is put in 'ON' position. Similarly in a bipolar junction transistor, by controlling base-emitter current, the emitter-collector resistance can be made either nearly infinite or nearly zero.

In a transistor characteristic, there are three regions. They are

- Cutoff Region
- Active Region
- Saturation Region

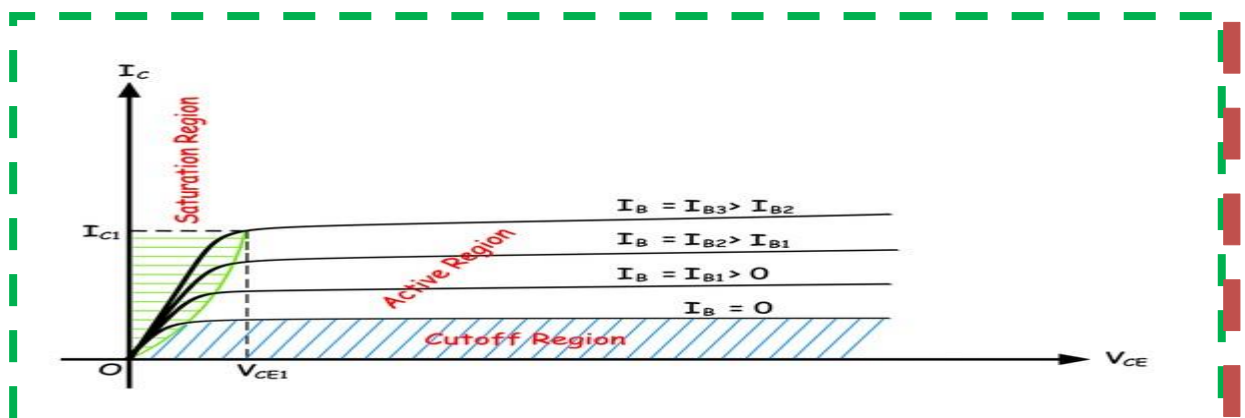
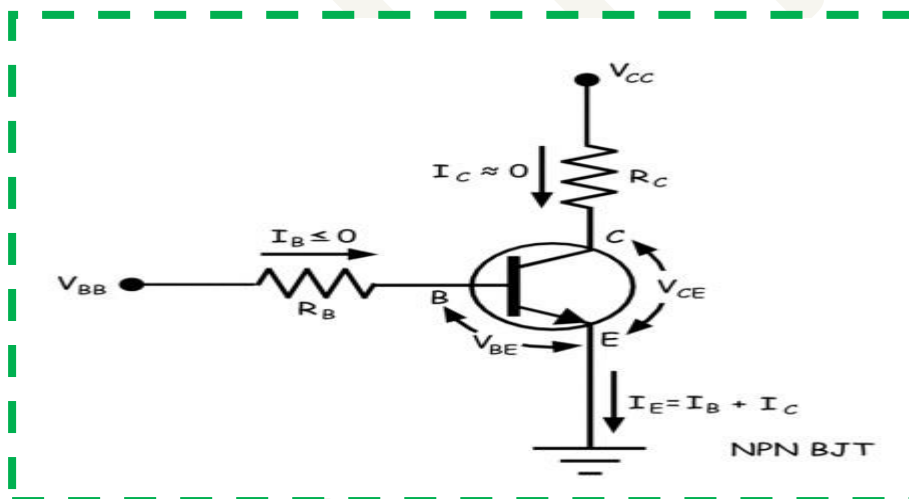


Figure 14: OUTPUT I-V CURVE OF BJT

Similarly, when the switch is ON, the voltage across the switch is zero hence, there is no power loss again. When we want a BJT to be operated as a switch, it must be operated in such a way that the power loss during ON and OFF condition should be nearly zero, or very low.

It is only possible when the transistor is only operated in the marginal region of the characteristics. The cutoff region and saturation region are two marginal regions in the transistor characteristics. Note that this applies to both npn transistors and pnp transistors.

In the figure, when base current is zero, the collector current (I_C) has very small constant value for a wide range of collector-emitter voltage (V_{CE}). So when the transistor is operated with base current ≤ 0 , the collector current ($I_C \approx 0$) is very tiny, hence the transistor is said to be in OFF condition but at the same time, power loss across the transistor switch i.e. $I_C \times V_{CE}$ is negligible because of very tiny I_C .



The transistor is connected in series with an output resistance R_C . Hence, current through the output resistance is

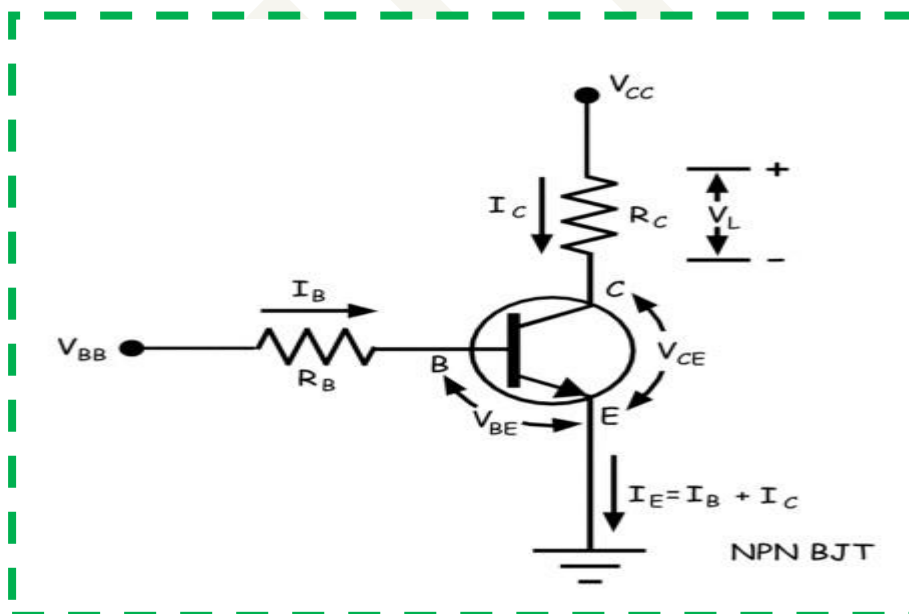
$$V_{CC} = I_C R_C + V_{CE}$$

$$\Rightarrow I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

$$\Rightarrow I_C = \frac{V_L}{R_C}$$

If the transistor is operated with a base current I_{B1} for which collector current is I_{C1} . If I_C is less than I_{C1} , then the transistor is operated in the saturation region. Here, for any collector current less than I_{C1} , there will be very tiny collector-emitter voltage ($V_{CE} < V_{CE1}$).

Hence in this situation, the current through the transistor is as high as load current, but the voltage across the transistor ($V_{CE} < V_{CE1}$) quite low, hence power loss in the transistor is negligible.



The transistor behaves as an ON switch. So for using the **transistor as a switch** we should make sure that the applied base current must be sufficiently high to keep the transistor in the saturation region, for a collector current.

LECTURE NO. 6

BJT CIRCUITS

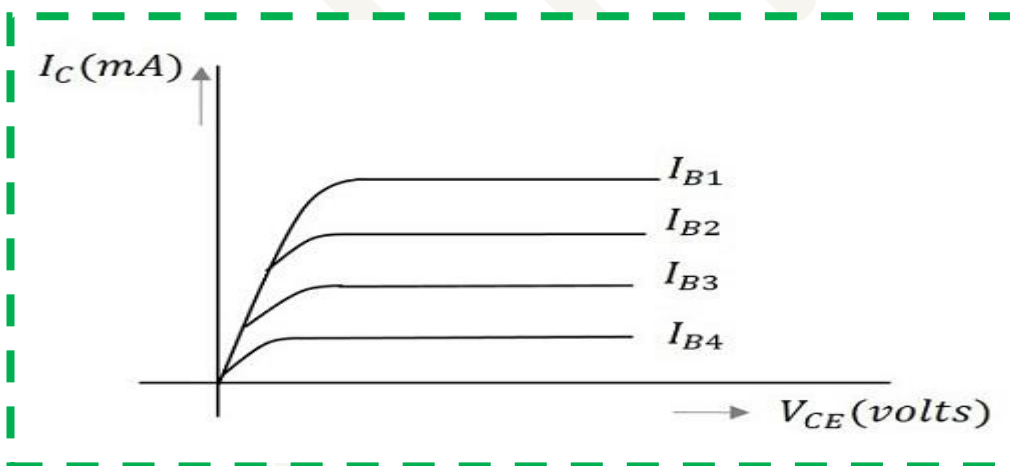
BJT BIASING

BJT BIASING CIRCUITS :

Transistor Biasing is the process of setting a transistors DC operating voltage or current conditions to the correct level so that any AC input signal can be amplified correctly by the transistor.

Output Characteristics

When the output characteristics of a transistor are considered, the curve looks as below for different input values.

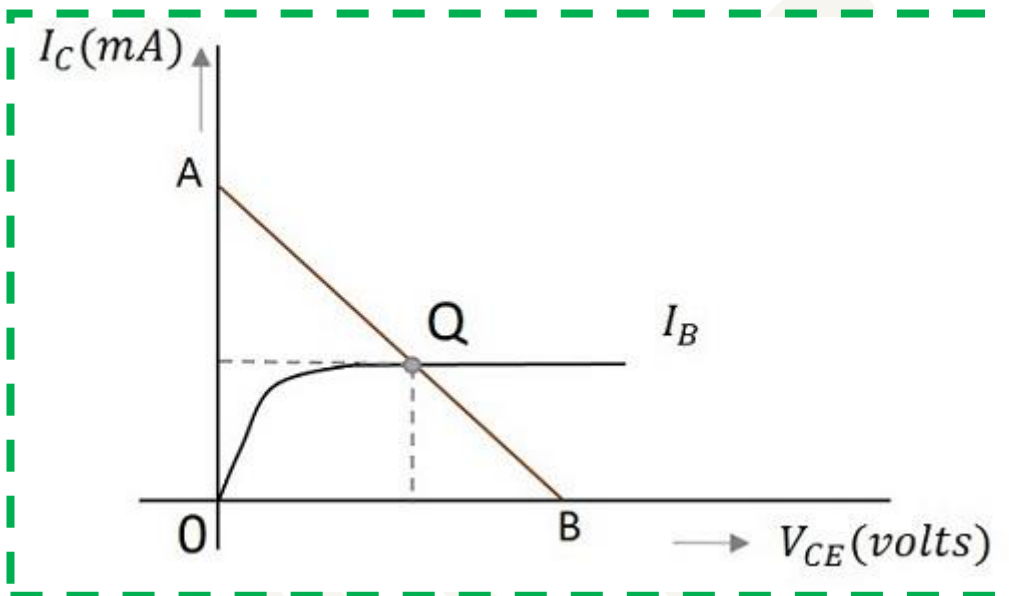


In the above figure, the output characteristics are drawn between collector current I_C and collector voltage V_{CE} for different values of base current I_B . These are considered here for different input values to obtain different output curves.

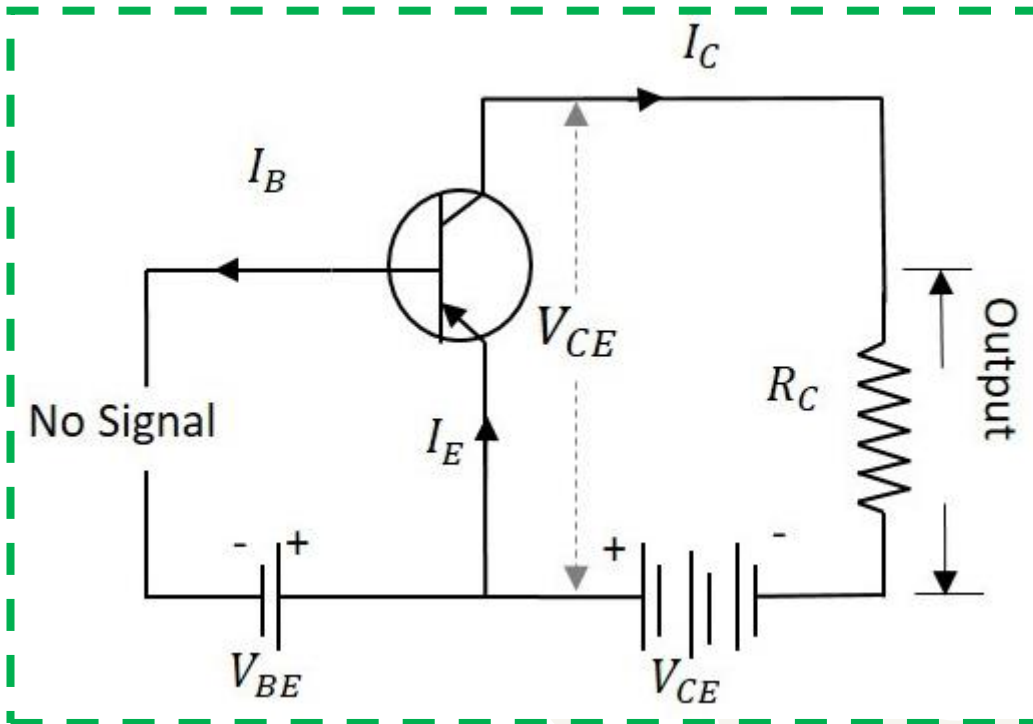
When a value for the maximum possible collector current is considered, that point will be present on the Y-axis, which is nothing but the saturation point. As well, when a value for the maximum possible collector emitter voltage is considered, that point will be present on the X-axis, which is the cutoff point.

When a line is drawn joining these two points, such a line can be called as Load line. This is called so as it symbolizes the output at the load. This line, when drawn over the output characteristic curve, makes contact at a point called as Operating point.

This operating point is also called as quiescent point or simply Q-point. There can be many such intersecting points, but the Q-point is selected in such a way that irrespective of AC signal swing, the transistor remains in active region. This can be better understood through the figure below.



When the transistor is given the bias and no signal is applied at its input, the load line drawn at such condition, can be understood as DC condition. Here there will be no amplification as the signal is absent. The circuit will be as shown below.



When collector emitter voltage $V_{CE} = 0$, the collector current is maximum and is equal to V_{CC}/R_C . This gives the maximum value of V_{CE} . This is shown as

$$V_{CE} = V_{CC} - I_C R_C$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = V_{CC}/R_C$$

When the collector current $I_C = 0$, then collector emitter voltage is maximum and will be equal to the V_{CC} . This gives the maximum value of I_C . This is shown as

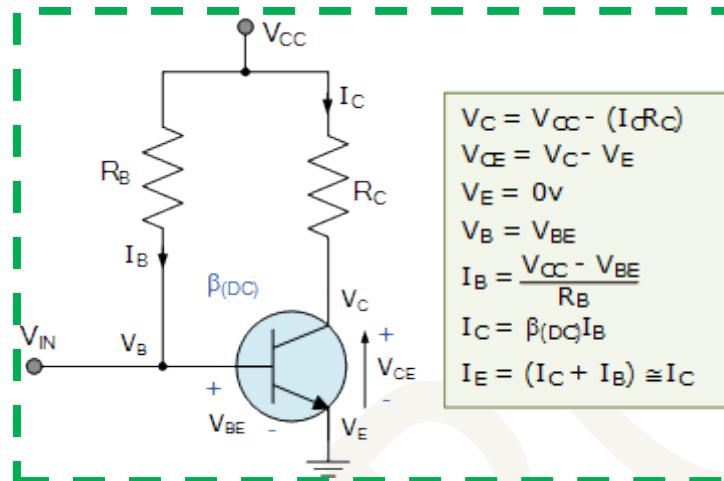
$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = V_{CC}$$

Establishing the correct operating point requires the selection of bias resistors and load resistors to provide the appropriate input current and collector voltage conditions. The correct biasing point for a bipolar transistor, either NPN or PNP, generally lies somewhere between the two extremes of operation with respect to it being either

“fully-ON” or “fully-OFF” along its DC load line. This central operating point is called the “Quiescent Operating Point”, or Q-point for short.

Fixed Base Biasing a Transistor



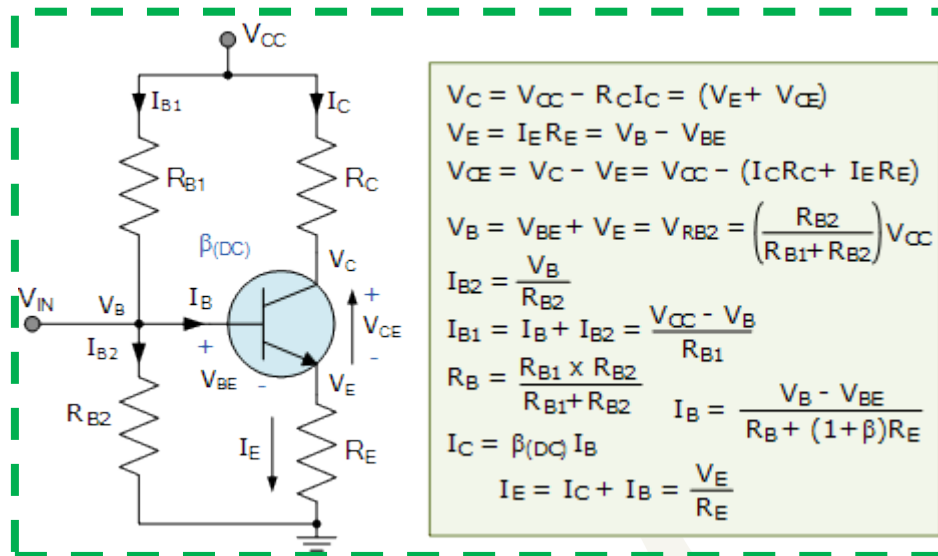
The circuit shown is called as a “fixed base bias circuit”, because the transistors base current, I_B remains constant for given values of V_{CC} , and therefore the transistors operating point must also remain fixed. This two resistor biasing network is used to establish the initial operating region of the transistor using a fixed current bias.

This type of transistor biasing arrangement is also beta dependent biasing as the steady-state condition of operation is a function of the transistors beta β value, so the biasing point will vary over a wide range for transistors of the same type as the characteristics of the transistors will not be exactly the same.

The emitter diode of the transistor is forward biased by applying the required positive base bias voltage via the current limiting resistor R_B . Assuming a standard bipolar transistor, the forward base-emitter voltage drop would be 0.7V. Then the value of R_B is simply: $(V_{CC} - V_{BE})/I_B$ where I_B is defined as I_C/β .

With this single resistor type of biasing arrangement the biasing voltages and currents do not remain stable during transistor operation and can vary enormously. Also the operating temperature of the transistor can adversely effect the operating point.

Voltage Divider Transistor Biasing



Here the common emitter transistor configuration is biased using a voltage divider network to increase stability. The name of this biasing configuration comes from the fact that the two resistors R_{B1} and R_{B2} form a voltage or potential divider network across the supply with their center point junction connected to the transistor's base terminal as shown.

This voltage divider biasing configuration is the most widely used transistor biasing method. The emitter diode of the transistor is forward biased by the voltage value developed across resistor R_{B2} . Also, voltage divider network biasing makes the transistor circuit independent of changes in beta as the biasing voltages set at the transistor's base, emitter, and collector terminals are not dependent on external circuit values.

To calculate the voltage developed across resistor R_{B2} and therefore the voltage applied to the base terminal we simply use the voltage divider formula for resistors in series.

Generally the voltage drop across resistor R_{B2} is much less than for resistor R_{B1} . Clearly the transistor's base voltage V_B with respect to ground, will be equal to the voltage across R_{B2} .

The amount of biasing current flowing through resistor R_{B2} is generally set to 10 times the value of the required base current I_B so that it is sufficiently high enough to have no effect on the voltage divider current or changes in Beta.

The goal of **Transistor Biasing** is to establish a known quiescent operating point, or Q-point for the bipolar transistor to work efficiently and produce an undistorted output signal. Correct DC biasing of the transistor also establishes its initial AC operating region with practical biasing circuits using either a two or four-resistor bias network.

In bipolar transistor circuits, the Q-point is represented by (V_{CE}, I_C) for the NPN transistors or (V_{EC}, I_C) for PNP transistors. The stability of the base bias network and therefore the Q-point is generally assessed by considering the collector current as a function of both Beta (β) and temperature.



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Name of Faculty: Parshant Tyagi

Subject & Subject Code: Basic Electrical Engineering, 1FY3-08

Semester: I

Session: 2020-21 (Odd Sem.)

Branch: First Year

Batch: A, B & C

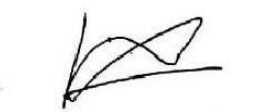
Unit: V

Date of Submission: 03/03/21

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V-I Characteristics of SCR

A V-I Characteristic of SCR (Silicon Controlled Rectifier) is the voltage current characteristics. The current through the SCR varies as the Anode to Cathode terminal voltage and Gate to Cathode terminal voltage is varied. The graphical representation of current through the SCR and voltage across the anode to cathode terminal is known as V-I Characteristics of SCR.

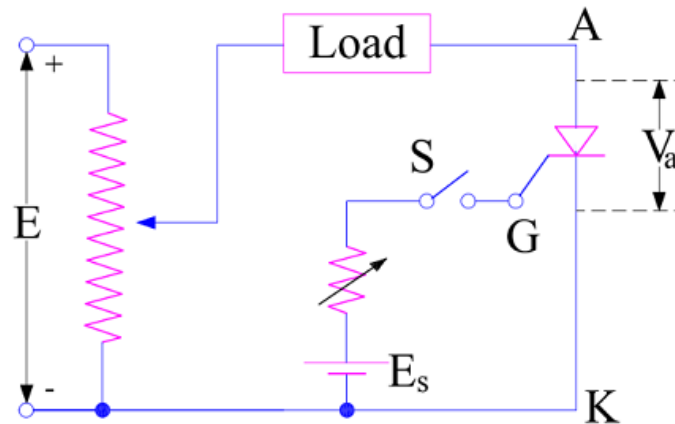
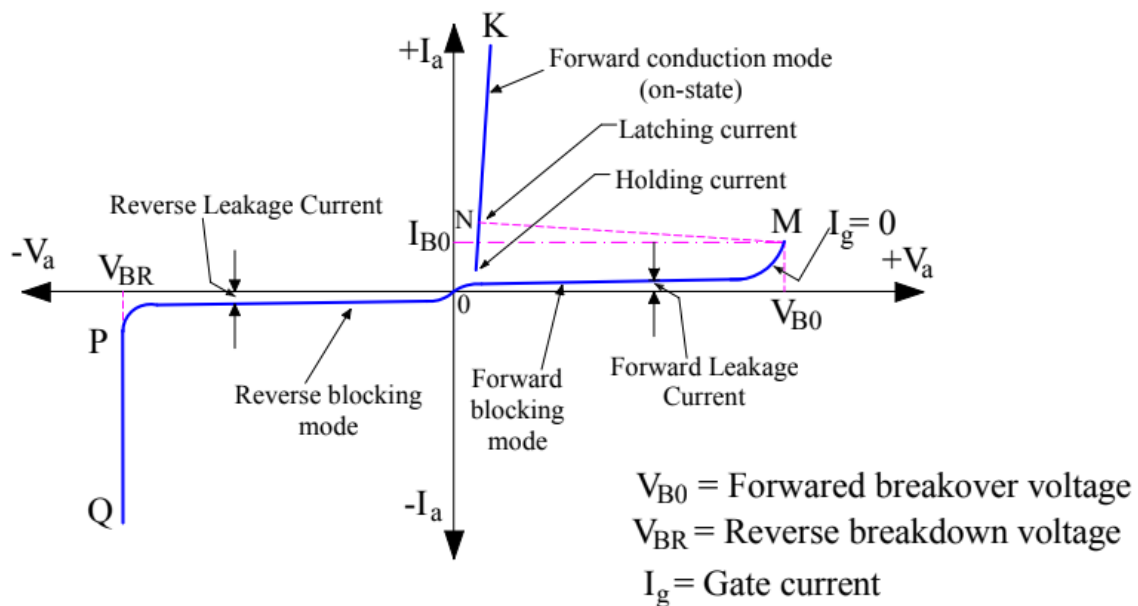


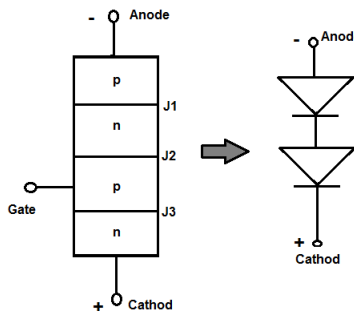
Fig:- Circuit Diagram for Obtaining V-I Characteristics of SCR

- V-I Characteristics of SCR is shown below



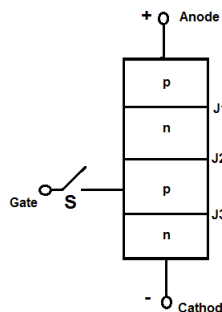
Various Modes in V-I Characteristics of SCR:-

Reverse Blocking Mode:- As clear from the V-I curve, the anode to cathode voltage is negative in this mode. This means that anode terminal is made negative with respect to cathode. Reverse Blocking Mode of SCR is that operational mode in which it offers high impedance for current flow and hence do not conduct. It is shown by OP in the V-I characteristics of SCR.



junction J1 and J3 are reversed biased while the junction J2 is forward biased. The device behaves as if two diodes are connected in series with reverse voltage applied across them.

Forward Blocking Mode:- Forward Blocking Mode is that operational mode of SCR in which it does not conduct even though it is forward biased. The term forward biased SCR implies that its anode terminal is positive with respect to cathode terminal with gate switch S open.



the junction J1 and J3 are forward biased but junction J2 is reverse biased. A small leakage current, called the forward leakage current, flows as shown by OM in the V-I characteristics of SCR in this mode.

Forward Conduction Mode:- In forward blocking mode, junction J1 and J3 are forward biased and J2 is reverse biased. This means, there are two possibilities for making SCR to conduct in this mode:

- Increase the anode to cathode voltage to such an extent which leads to avalanche breakdown of the reverse biased junction J2.

- Apply positive gate pulse between gate and cathode terminal.

the anode current in this mode will only be limited by the load, so based on the value of load the anode current will change and may lie at any point in between N and K. Thus NK represents the forward conduction of SCR.

- Important Terms In The V-I Characteristics of SCR

Breakover voltage :- It is the minimum forward voltage, gate being open, at which SCR starts conducting heavily i.e. turned on.

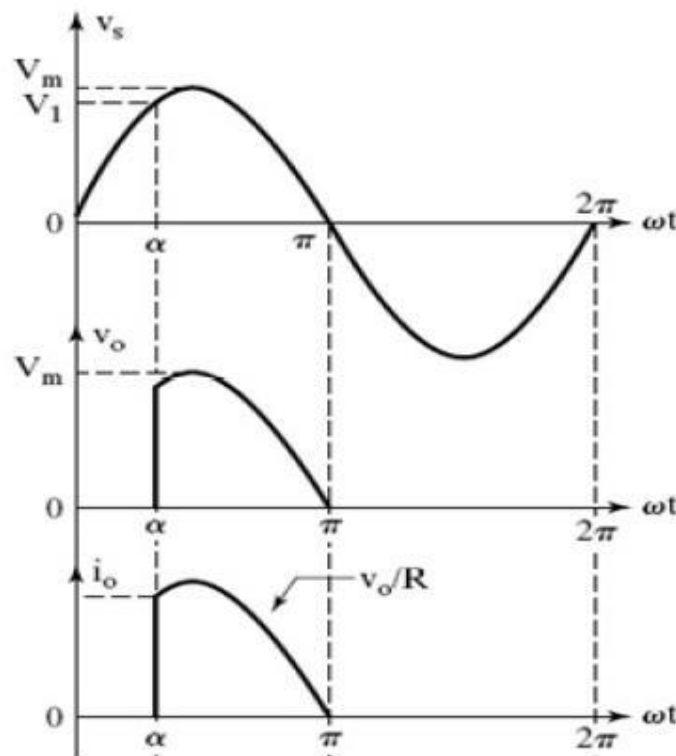
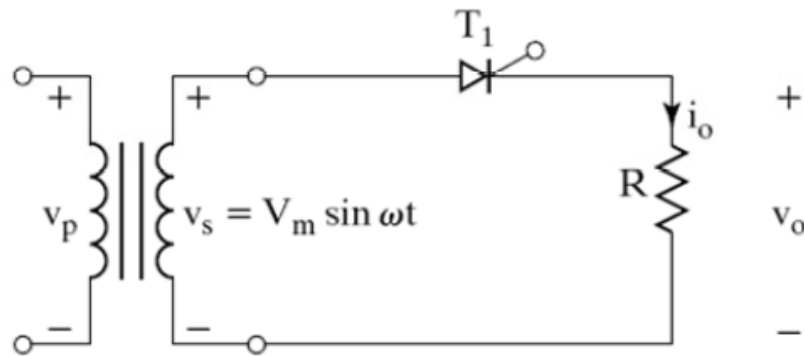
Peak reverse voltage :- It is the maximum reverse voltage (cathode positive w.r.t. anode) that can be applied to an SCR without conducting in the reverse direction.

Holding current :- Holding current in thyristor or SCR can be defined as, the smallest amount of current under which anode current has to drop to enter OFF status.

Latching current :- The latching current is the smallest amount of anode current is required for preserving the thyristor in the ON condition instantly once a thyristor is turned ON then the gate signal has been detached.

Single Phase Half Wave Controlled Rectifier

A **half wave rectifier** is defined as a type of rectifier that only allows one half-cycle of an AC voltage waveform to pass, blocking the other half-cycle.



The circuit using an SCR as an half-wave rectifier is shown in figure. The ac supply to be rectified is applied to the primary of the transformer ensuring that the negative voltage appearing at the secondary of the transformer is less than reverse breakdown voltage of the SCR. The load resistance R_L is connected in series with anode. A variable resistance r is inserted in the gate circuit for control of gate current. the angle at which the SCR starts conducting (i.e. firing angle)

$$\text{Average output voltage, } V_{av} = V_{MAX}/2\pi (1 + \cos \alpha)$$

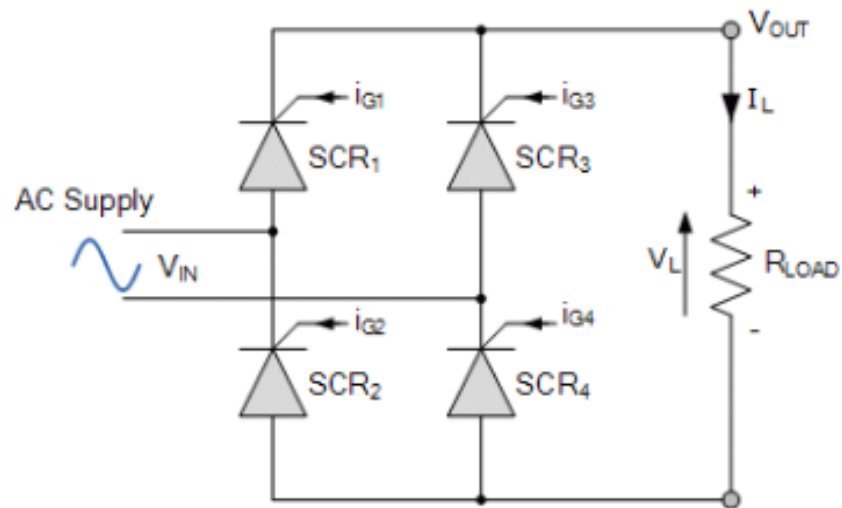
$$\text{Average current, } I_{av} = V_{MAX}/2\pi R_L (1 + \cos \alpha)$$

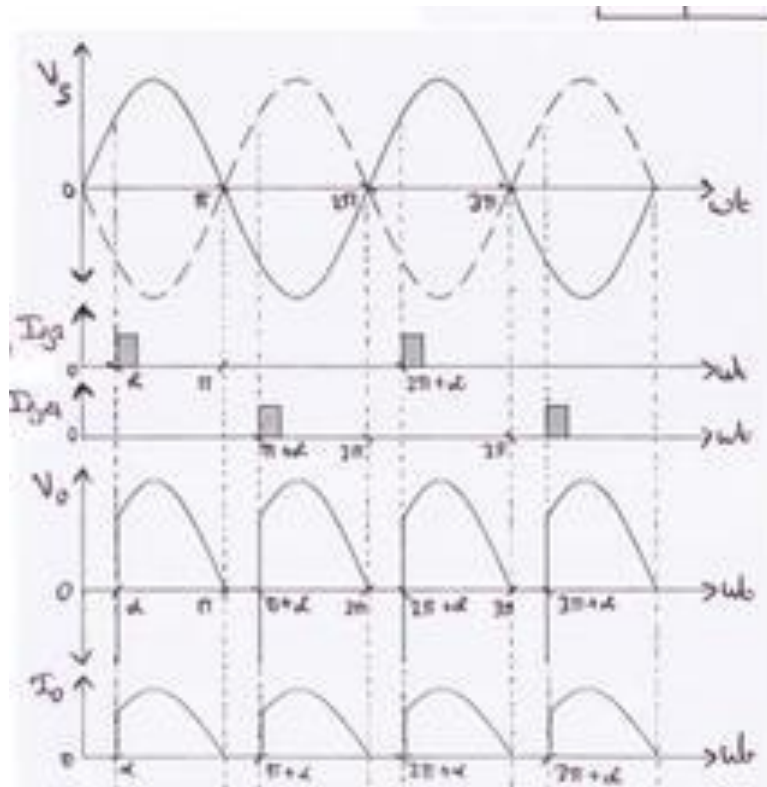
Thus the desired value of average current, I_{av} can be obtained by varying firing angle α .

$$I_{av} = V_{MAX}/\pi R_L \text{ when } \alpha = 0$$

$$I_{av} = V_{MAX}/2\pi R_L \text{ when } \alpha = \pi/2$$

Single Phase Full Wave Controlled Rectifier

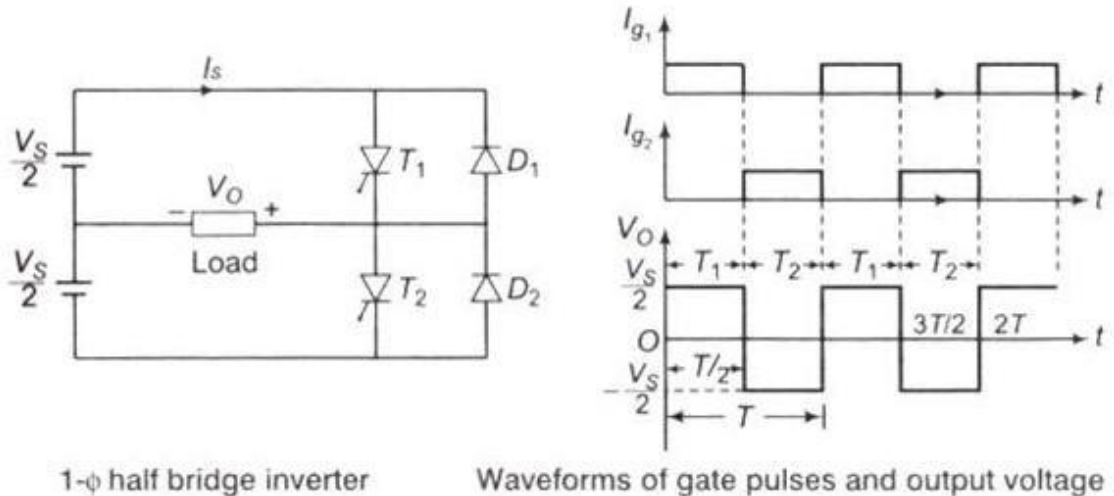




Single Phase Half Bridge Inverter

Half-bridge inverter requires two diodes and two switches which are connected in anti-parallel. The two switches are complementary switches which means when the first switch is ON the second switch will be OFF. Similarly, when the second switch is ON the first switch will be OFF.

Circuit Diagram:-



Case 1 (when switch S_1 is ON and S_2 is OFF): When switch S_1 is ON from a time period of 0 to $T/2$, the diode D_1 and D_2 are in reverse bias condition and S_2 switch is OFF.

Applying KVL (Kirchhoff's Voltage Law)

$$V_s/2 - V_o = 0$$

Where output voltage $V_o = V_s/2$

Case 2 (when switch S_2 is ON and S_1 is OFF): When switch S_2 is ON from a time period of $T/2$ to T , the diode D_1 and D_2 are in reverse bias condition and S_1 switch is OFF.

Applying KVL (Kirchhoff's Voltage Law)

$$V_s/2 + V_0 = 0$$

Where output voltage $V_0 = -V_s/2$

Where output current $i_0 = V_0/R = -V_s/2R$

In case of supply current or switch current, the current $i_{S1} = 0$, $i_{S2} = i_0 = -V_s/2R$ and the diode current $i_{D1} = i_{D2} = 0$.

The average value of output voltage is

$$V_{0(avg)} = \frac{1}{T} \int_0^T V_{0(\omega t)} d\omega t$$

Using Fourier series

$$V_{0(\omega t)} = V_{0(avg)} + \sum_{n=1}^{\infty} C_n \sin(n\omega t + \phi_n)$$

Where C_n , a_n and b_n are

$$C_n = \sqrt{a_n^2} + \sqrt{b_n^2}$$

$$a_n = \frac{2}{T} \int_0^T V_{0(\omega t)} \cos n\omega t d\omega t = 0$$

$$b_n = V_s/n\pi(1 - \cos n\pi)$$

The $b_n = 0$ when substituting even numbers ($n=2,4,6,\dots$) and $b_n = 2V_s/n\pi$ when substituting odd numbers ($n=1,3,5,\dots$). Substitute $b_n = 2V_s/n\pi$ and $a_n=0$ in C_n will get $C_n = 2V_s/n\pi$.

The fundamental voltage can be written as

$$V_{01(\omega t)} = 2V_s/n * (\sin \omega t)$$

The maximum value of fundamental voltage

$$V_{01(\max)} = 2V_s/n$$

The RMS value of the fundamental voltage is

$$V_{01(RMS)} = 2V_s/\sqrt{2n} = \sqrt{2}V_s/n$$

The fundamental component of the RMS output current is

$$I_{01(RMS)} = V_{01(RMS)}/R$$

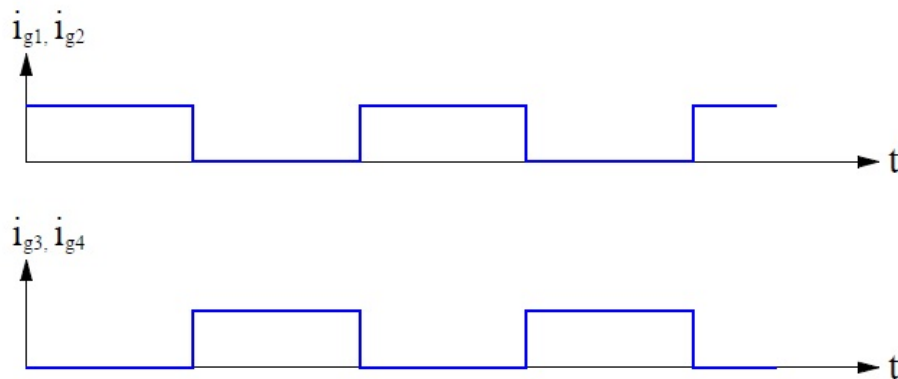
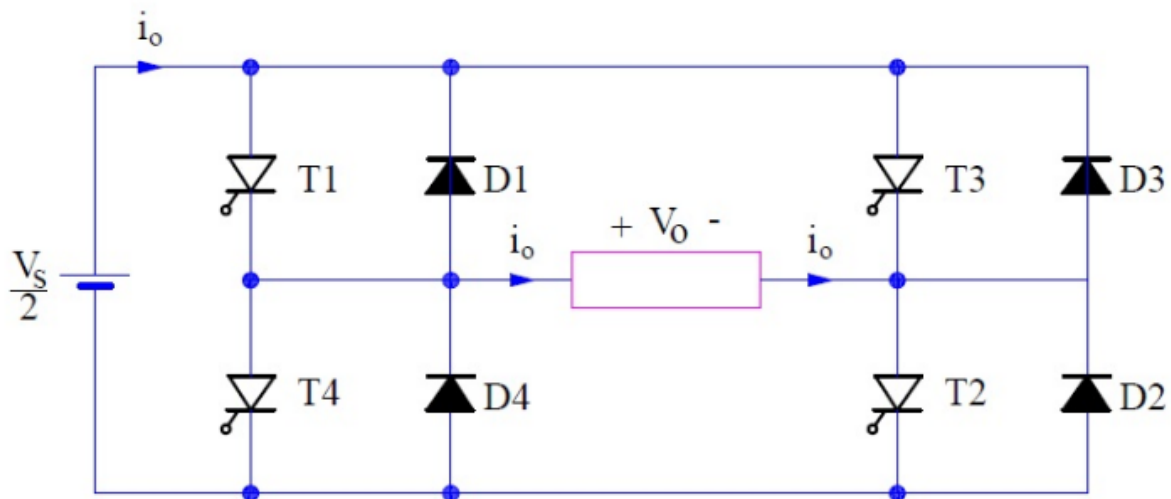
The total **harmonic distortion** is expressed as

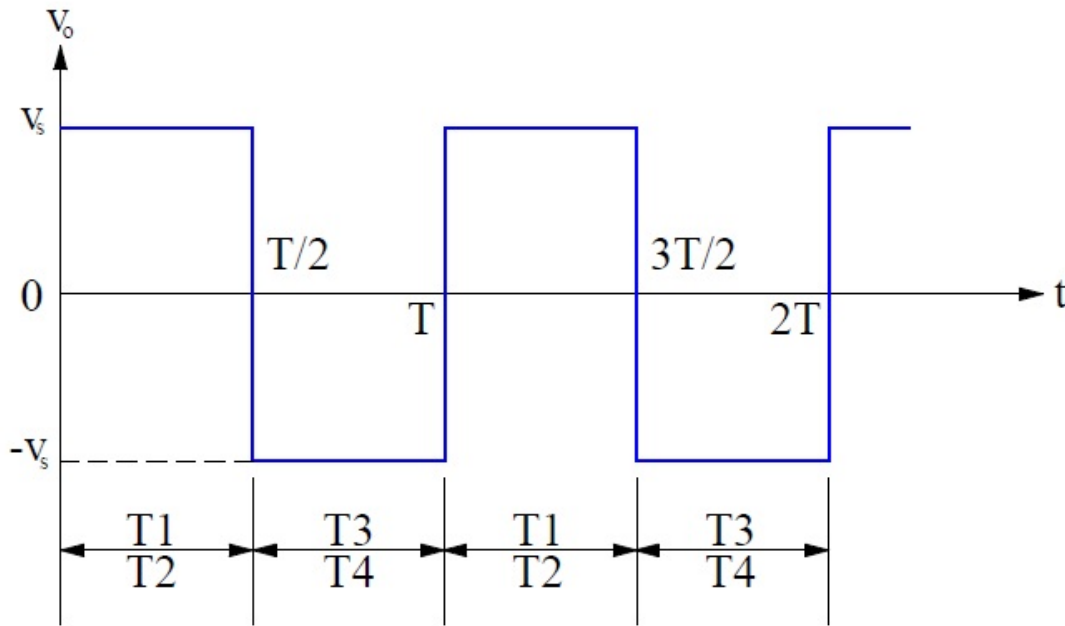
$$THD = \sqrt{\frac{1}{g^2} - 1} = \sqrt{\frac{\pi^2}{(2\sqrt{2})^2}}$$

In the output voltage the total harmonic distortion THD=48.43%

Circuit Diagram of Single Phase Full Bridge Inverter:

The power circuit of a single phase full bridge inverter comprises of four thyristors T1 to T4, four diodes D1 to D4 and a two wire DC input power source V_s . Each diode is connected in antiparallel to the thyristors viz. D1 is connected in anti-parallel to T1 and so on. The power circuit diagram of a single phase full bridge inverter is shown in the figure below.





For the time $0 \leq t < T/2$, thyristors T1 & T2 conducts and load voltage $V_o = V_s$.

For the time $(T/2) \leq t < T$, thyristors T3 & T4 conducts and load voltage $V_o = -V_s$.

Purpose of Diodes D1 to D4:

If the load is purely resistive, there is no need to put diode D1 to D4 as the output voltage and current are always in phase. But unfortunately, for loads other than purely resistive, the load current (i_o) will not be in phase with the load voltage (v_o). For such case, the diode connected in anti-parallel with the thyristor will allow the current to flow when main thyristor is turned off. When these diode conducts, the energy is fed back to the DC source and hence, these diodes (D1 to D4) are called [flyback diode](#).

$$V_{0(rms)} = \left[\frac{1}{T} \left(\int_0^{T/2} V_s^2 dt + \int_{T/2}^T V_s^2 dt \right) \right]^{\frac{1}{2}}$$

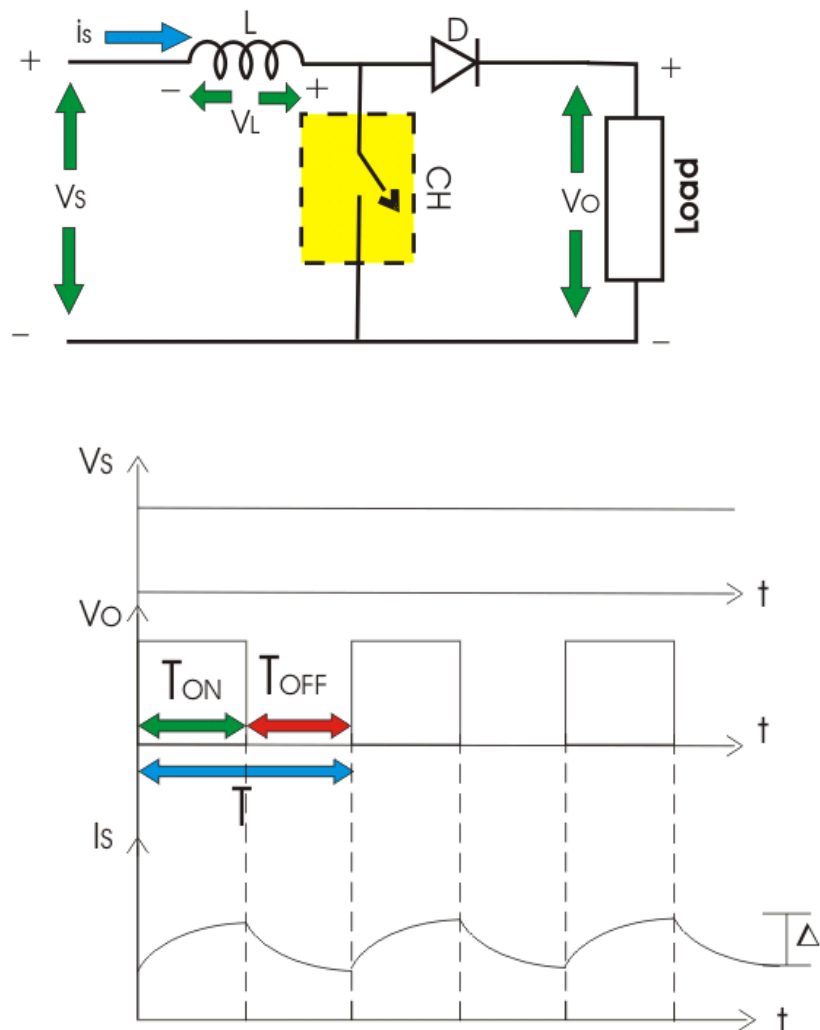
$$V_0 = \sum_{n=1,3,5}^{\infty} \left(\frac{4V_s}{n\pi} \right) \sin(n\omega t)$$

DC- DC Converter

Chopper is a basically static power electronics device which converts fixed DC voltage/power to variable DC voltage or power. It is nothing but a high speed switch which connects and disconnects the load from source at a high rate to get variable or chopped voltage at the output.

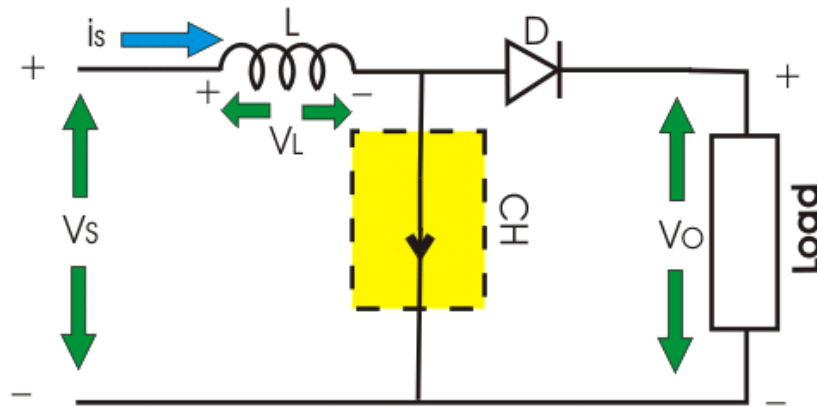
Step up Chopper or Boost Converter :

Step up chopper or boost converter is used to increase the input voltage level of its output side. Its circuit diagram and waveforms are shown below in figure.



When CH is ON it short circuits the load. Hence output voltage during T_{ON} is zero. During this period inductor gets charged. So, $V_s = V_L$

$$L \frac{di}{dt} = V_s \Rightarrow \frac{\Delta I}{T_{ON}} = \frac{V_s}{L} \Rightarrow \Delta I = \frac{V_s}{L} T_{ON}$$

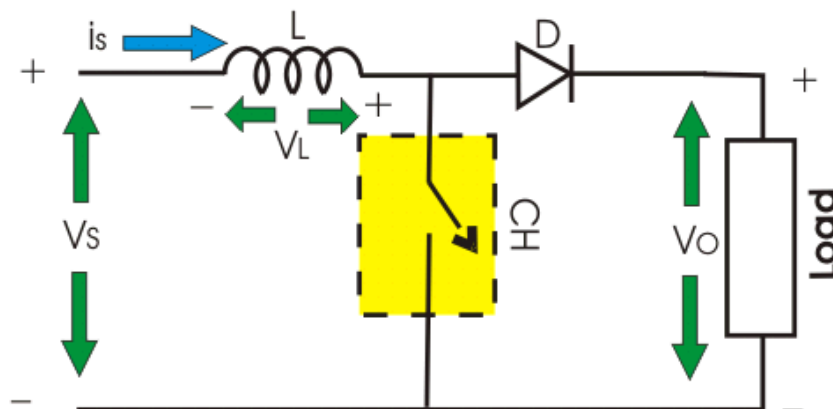


Where, ΔI is the peak to peak inductor current.

When CH is OFF inductor L discharges through the load. So, we will get summation of both source voltage V_s and inductor Voltage V_L as output voltage, i.e.

$$V_o = V_s + V_L \Rightarrow V_L = V_o - V_s \Rightarrow L \frac{di}{dt} = V_o - V_s$$

$$L \frac{\Delta I}{T_{OFF}} = V_o - V_s \Rightarrow \Delta I = \frac{V_o - V_s}{L} T_{OFF}$$



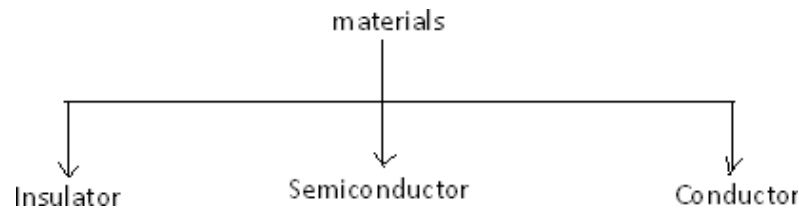
$$\text{average output voltage, } V_o = \frac{V_s}{1-D}$$

As we can vary TON from 0 to T, so $0 \leq D \leq 1$. Hence V_o can be varied from V_s to ∞ . It is clear that output voltage is always greater than the input voltage and hence it boost up or increase the voltage level.

DIODE CIRCUITS (SEMICONDUCTOR)

INTRODUCTION

Based on the electrical conductivity all the materials in nature are classified as insulators, semiconductors, and conductors.



Insulator: An insulator is a material that offers a very low level (or negligible) of conductivity when voltage is applied. Eg: Paper, Mica, glass, quartz. Typical resistivity level of an insulator is of the order of 10^{10} to $10^{12} \Omega\text{-cm}$. The energy band structure of an insulator is shown in the fig.1.1. Band structure of a material defines the band of energy levels that an electron can occupy. Valance band is the range of electron energy where the electron remain bended too the atom and do not contribute to the electric current. Conduction bend is the range of electron energies higher than valance band where electrons are free to accelerate under the influence of external voltage source resulting in the flow of charge.

The energy band between the valance band and conduction band is called as forbidden band gap. It is the energy required by an electron to move from balance band to conduction band i.e. the energy required for a valance electron to become a free electron.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

For an insulator, as shown in the fig.1.1 there is a large forbidden band gap of greater than 5eV. Because of this large gap there a very few electrons in the CB and hence the conductivity of insulator is poor. Even an increase in temperature or applied electric field is insufficient to transfer electrons from VB to CB.

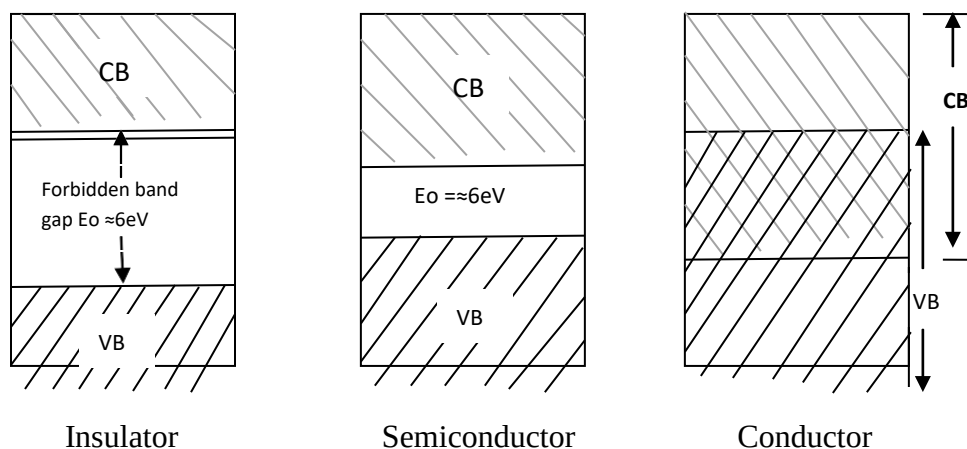


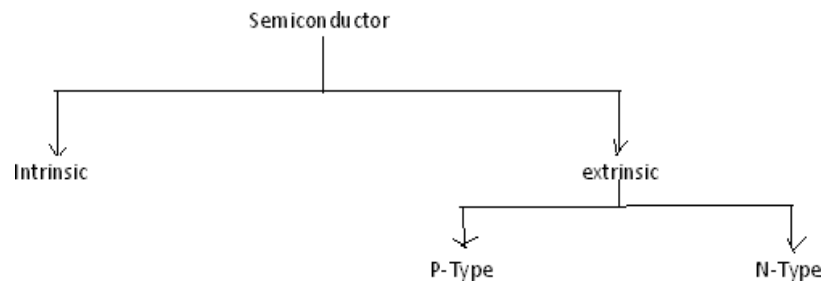
FIG:1.1 Energy band diagrams insulator, semiconductor and conductor

Conductors: A conductor is a material which supports a generous flow of charge when a voltage is applied across its terminals. i.e. it has very high conductivity. Eg: Copper, Aluminum, Silver, Gold. The resistivity of a conductor is in the order of 10^{-4} and $10^{-6} \Omega\text{-cm}$. The Valance and conduction bands overlap (fig1.1) and there is no energy gap for the electrons to move from valance band to conduction band. This implies that there are free electrons in CB even at absolute zero temperature (0K). Therefore at room temperature when electric field is applied large current flows through the conductor.

Semiconductor: A semiconductor is a material that has its conductivity somewhere between the insulator and conductor. The resistivity level is in the range of 10 and $10^4 \Omega\text{-cm}$. Two of the most commonly used are Silicon (Si=14 atomic no.) and germanium (Ge=32 atomic no.). Both have 4 valance electrons. The forbidden band gap is in the order of 1eV. For eg., the band gap energy for Si, Ge and GaAs is 1.21, 0.785 and 1.42 eV, respectively at absolute zero temperature (0K). At 0K and at low temperatures, the valance band electrons do not have sufficient energy to move from V to CB. Thus semiconductors act as insulators at 0K. as the temperature increases, a large number of valance electrons acquire sufficient energy to leave the VB, cross the forbidden bandgap and reach CB. These are now free electrons as they can move freely under the influence of electric field. At room temperature there are sufficient electrons in the CB and hence the semiconductor is capable of conducting some current at room temperature.

Inversely related to the conductivity of a material is its resistance to the flow of charge or current. Typical resistivity values for various materials' are given as follows.

Semiconductor Types



A pure form of semiconductors is called as intrinsic semiconductor. Conduction in intrinsic sc is either due to thermal excitation or crystal defects. Si and Ge are the two most important semiconductors used. Other examples include Gallium arsenide GaAs, Indium Antimonide (InSb) etc.

Let us consider the structure of Si. A Si atomic no. is 14 and it has 4 valance electrons. These 4 electrons are shared by four neighboring atoms in the crystal structure by means of covalent bond. Fig. 1.2a shows the crystal structure of Si at absolute zero temperature (0K). Hence a pure SC acts has poor conductivity (due to lack of free electrons) at low or absolute zero temperature.

The absence of electrons in covalent bond is represented by a small circle usually referred to as hole which is of positive charge. Even a hole serves as carrier of electricity in a manner similar to that of free electron. In a pure semiconductor, the number of holes is equal to the number of free electrons.

EXTRINSIC SEMICONDUCTOR

Intrinsic semiconductor has very limited applications as they conduct very small amounts of current at room temperature. The current conduction capability of intrinsic semiconductor can be increased significantly by adding a small amount of impurity to the intrinsic semiconductor. By adding impurities it becomes impure or extrinsic semiconductor. This process of adding impurities is called as doping. The amount of impurity added is 1 part in 10^6 atoms.

N type semiconductor: If the added impurity is a pentavalent atom then the resultant semiconductor is called N-type semiconductor. Examples of pentavalent impurities are Phosphorus, Arsenic, Bismuth, Antimony etc.

P type semiconductor: If the added impurity is a trivalent atom then the resultant semiconductor is called P-type semiconductor. Examples of trivalent impurities are Boron, Gallium, Indium etc. Thus in P type sc, holes are majority carriers and electrons are minority carriers. Since each trivalent impurity atom is capable of accepting an electron, these are called as acceptor atoms. The following fig 1.5b shows the pictorial representation of P type sc

- The conductivity of N type sc is greater than that of P type sc as the mobility of electron is greater than that of hole.
- For the same level of doping in N type sc and P type sc, the conductivity of an N type sc is around twice that of a P type sc.

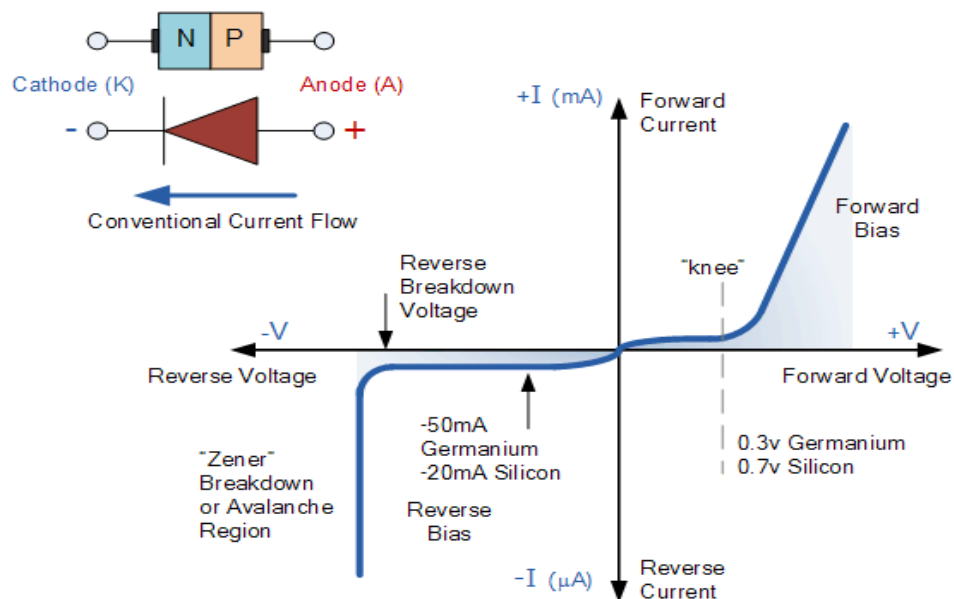
A *PN Junction Diode* is one of the simplest semiconductor devices around, and which has the characteristic of passing current in only one direction only. However, unlike a resistor, a diode does not behave linearly with respect to the applied voltage as the diode has an exponential current-voltage (I-V) relationship and therefore we cannot describe its operation by simply using an equation such as Ohm's law.

If a suitable positive voltage (forward bias) is applied between the two ends of the PN junction, it can supply free electrons and holes with the extra energy they require to cross the junction as the width of the depletion layer around the PN junction is decreased.

By applying a negative voltage (reverse bias) results in the free charges being pulled away from the junction resulting in the depletion layer width being increased. This has the effect of increasing or decreasing the effective resistance of the junction itself allowing or blocking current flow through the diode.

Then the depletion layer widens with an increase in the application of a reverse voltage and narrows with an increase in the application of a forward voltage. This is due to the differences in the electrical properties on the two sides of the PN junction resulting in physical changes taking place. One of the results produces rectification as seen in the PN junction diodes static I-V (current-voltage) characteristics. Rectification is shown by an asymmetrical current flow when the polarity of bias voltage is altered as shown below.

Junction Diode Symbol and Static I-V Characteristics



But before we can use the PN junction as a practical device or as a rectifying device we need to firstly bias the junction, i.e. connect a voltage potential across it. On the voltage axis above, “Reverse Bias” refers to an external voltage potential which increases the potential barrier. An external voltage which decreases the potential barrier is said to act in the “Forward Bias” direction.

There are two operating regions and three possible “biasing” conditions for the standard **Junction Diode** and these are:

1. Zero Bias – No external voltage potential is applied to the PN junction diode.
2. Reverse Bias – The voltage potential is connected negative, (-ve) to the P-type material and positive, (+ve) to the N-type material across the diode which has the effect of Increasing the PN junction diode’s width.
3. Forward Bias – The voltage potential is connected positive, (+ve) to the P-type material and negative, (-ve) to the N-type material across the diode which has the effect of Decreasing the PN junction diodes width.

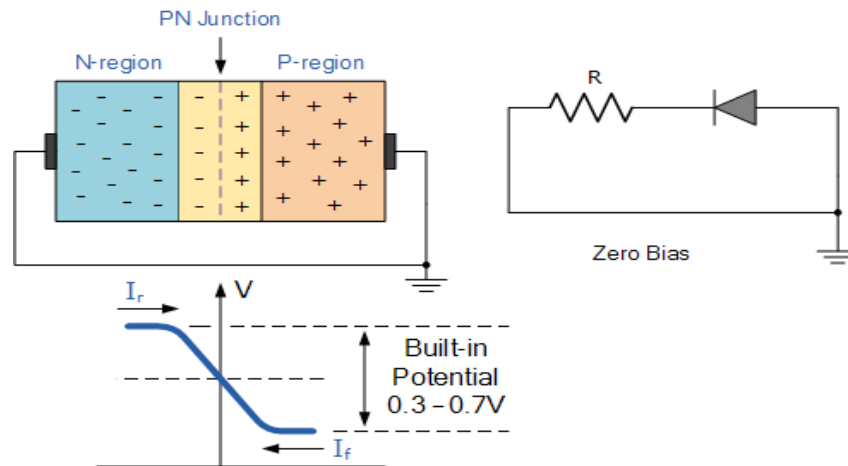
Zero Biased Junction Diode

When a diode is connected in a Zero Bias condition, no external potential energy is applied to the PN junction. However if the diodes terminals are shorted together, a few holes (majority carriers) in the P-type material with enough energy to overcome the potential barrier will move across the junction against this barrier potential. This is known as the “Forward Current” and is referenced as I_F

Likewise, holes generated in the N-type material (minority carriers), find this situation favourable and move across the junction in the opposite direction. This is known as the “Reverse Current” and is referenced as I_R . This transfer of electrons and holes back and forth across the PN junction is known as diffusion, as shown below.

LECTURE-2 PN JUNCTION DIODE

Zero Biased PN Junction Diode



The potential barrier that now exists discourages the diffusion of any more majority carriers across the junction. However, the potential barrier helps minority carriers (few free electrons in the P-region and few holes in the N-region) to drift across the junction.

The minority carriers are constantly generated due to thermal energy so this state of equilibrium can be broken by raising the temperature of the PN junction causing an increase in the generation of minority carriers, thereby resulting in an increase in leakage current but an electric current cannot flow since no circuit has been connected to the PN junction.

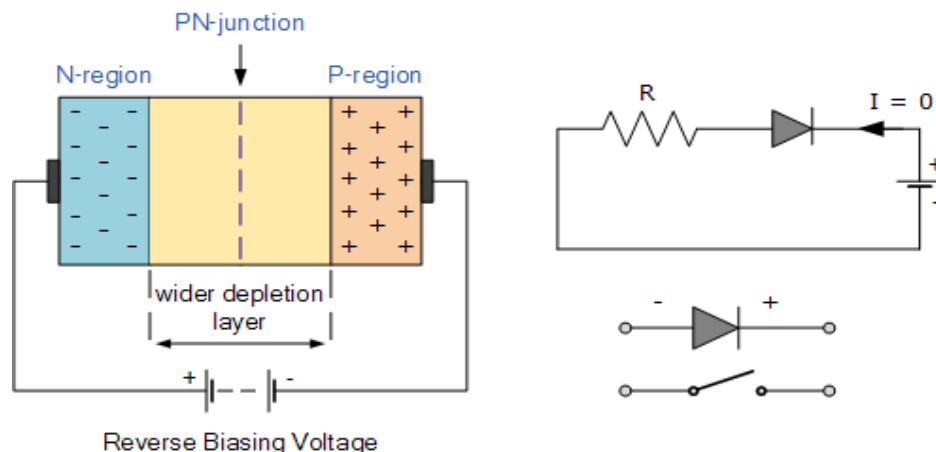
Reverse Biased PN Junction Diode

When a diode is connected in a Reverse Bias condition, a positive voltage is applied to the N-type material and a negative voltage is applied to the P-type material.

The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the P-type end are also attracted away from the junction towards the negative electrode.

The net result is that the depletion layer grows wider due to a lack of electrons and holes and presents a high impedance path, almost an insulator. The result is that a high potential barrier is created thus preventing current from flowing through the semiconductor material.

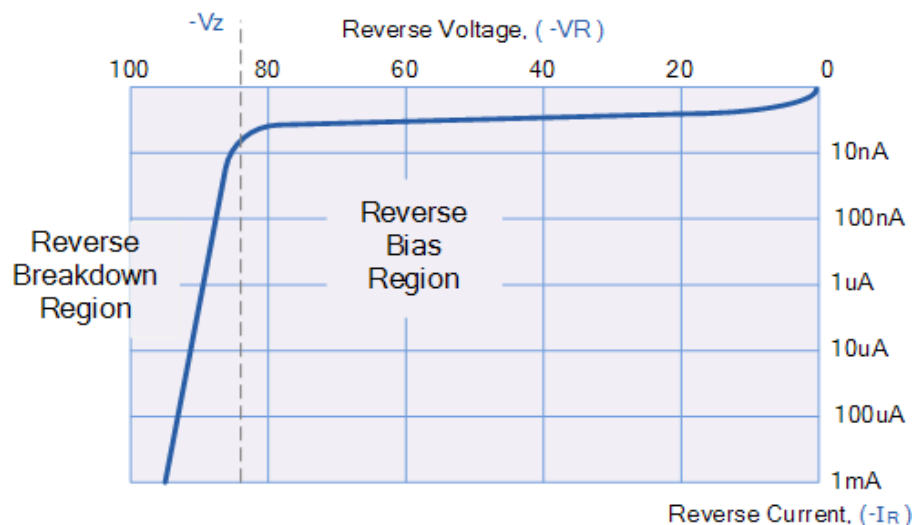
Increase in the Depletion Layer due to Reverse Bias



This condition represents a high resistance value to the PN junction and practically zero current flows through the junction diode with an increase in bias voltage. However, a very small leakage current does flow through the junction which can be measured in micro-amperes, (μA).

One final point, if the reverse bias voltage V_r applied to the diode is increased to a sufficiently high enough value, it will cause the diode's PN junction to overheat and fail due to the avalanche effect around the junction. This may cause the diode to become shorted and will result in the flow of maximum circuit current, and this shown as a step downward slope in the reverse static characteristics curve below.

Reverse Characteristics Curve for a Junction Diode



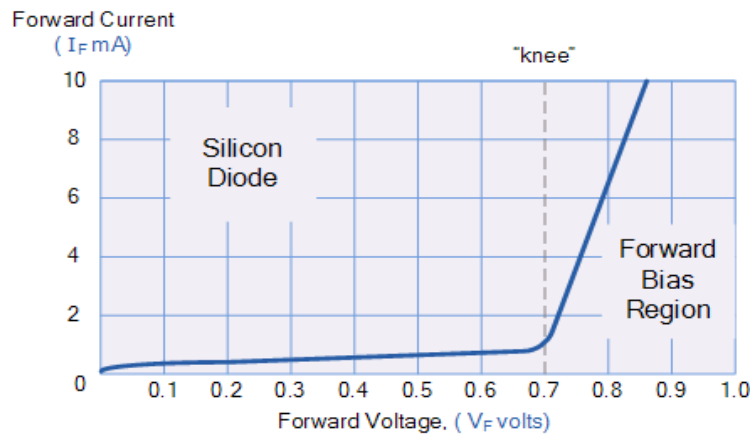
Sometimes this avalanche effect has practical applications in voltage stabilizing circuits where a series limiting resistor is used with the diode to limit this reverse breakdown current to a preset maximum value thereby producing a fixed voltage output across the diode. These types of diodes are commonly known as Zener Diodes and are discussed in a later tutorial.

Forward Biased PN Junction Diode

When a diode is connected in a Forward Bias condition, a negative voltage is applied to the N-type material and a positive voltage is applied to the P-type material. If this external voltage becomes greater than the value of the potential barrier, approx. 0.7 volts for silicon and 0.3 volts for germanium, the potential barriers opposition will be overcome and current will start to flow.

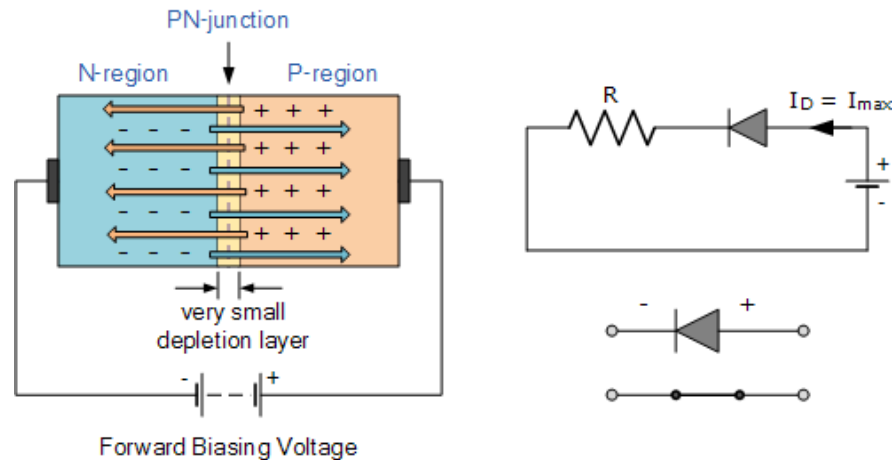
This is because the negative voltage pushes or repels electrons towards the junction giving them the energy to cross over and combine with the holes being pushed in the opposite direction towards the junction by the positive voltage. This results in a characteristics curve of zero current flowing up to this voltage point, called the “knee” on the static curves and then a high current flow through the diode with little increase in the external voltage as shown below.

Forward Characteristics Curve for a Junction Diode



The application of a forward biasing voltage on the junction diode results in the depletion layer becoming very thin and narrow which represents a low impedance path through the junction thereby allowing high currents to flow. The point at which this sudden increase in current takes place is represented on the static I-V characteristics curve above as the “knee” point.

Reduction in the Depletion Layer due to Forward Bias



This condition represents the low resistance path through the PN junction allowing very large currents to flow through the diode with only a small increase in bias voltage. The actual potential difference across the junction or diode is kept constant by the action of the depletion layer at approximately 0.3v for germanium and approximately 0.7v for silicon junction diodes.

Since the diode can conduct “infinite” current above this knee point as it effectively becomes a short circuit, therefore resistors are used in series with the diode to limit its current flow. Exceeding its maximum forward current specification causes the device to dissipate more power in the form of heat than it was designed for resulting in a very quick failure of the device.

Junction Diode Summary

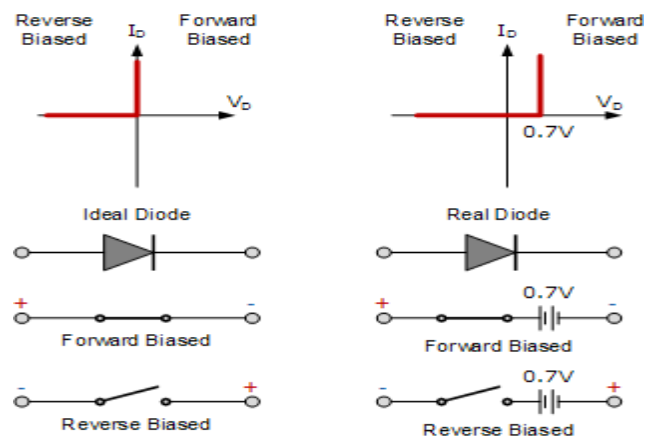
- The PN junction region of a **Junction Diode** has the following important characteristics:
- Semiconductors contain two types of mobile charge carriers, “Holes” and “Electrons”.
- The holes are positively charged while the electrons negatively charged.
- A semiconductor may be doped with donor impurities such as Antimony (N-type doping), so that it contains mobile charges which are primarily electrons.
- A semiconductor may be doped with acceptor impurities such as Boron (P-type doping),

so that it contains mobile charges which are mainly holes.

- The junction region itself has no charge carriers and is known as the depletion region.
- The junction (depletion) region has a physical thickness that varies with the applied voltage.
- When a diode is Zero Biased no external energy source is applied and a natural Potential Barrier is developed across a depletion layer which is approximately 0.5 to 0.7v for silicon diodes and approximately 0.3 of a volt for germanium diodes.
- When a junction diode is Forward Biased the thickness of the depletion region reduces and the diode acts like a short circuit allowing full current to flow.
- When a junction diode is Reverse Biased the thickness of the depletion region increases and the diode acts like an open circuit blocking any current flow, (only a very small leakage current).

We have also seen above that the diode is two terminal non-linear device whose I-V characteristic are polarity dependent as depending upon the polarity of the applied voltage, V_D the diode is either *Forward Biased*, $V_D > 0$ or *Reverse Biased*, $V_D < 0$. Either way we can model these current-voltage characteristics for both an ideal diode and for a real silicon diode as shown:

Junction Diode Ideal and Real Characteristics



For the operation of most of the electronics devices and circuits, a d.c. source is required. So it is advantageous to convert domestic a.c. supply into d.c voltages. The process of converting a.c. voltage into d.c. voltage is called as rectification. This is achieved with i) Step-down Transformer, ii) Rectifier, iii) Filter and iv) Voltage regulator circuits.

These elements constitute d.c. regulated power supply shown in the fig 1 below.